Tariff Shocks, Household Exposure, and Aggregate Demand*

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Abstract

How does household heterogeneity shape the macroeconomic effects of tariffs? I show that the direct expenditure response depends not only on average import exposure, but also on whether exposure is concentrated among households with high spending propensities. I derive a sufficient statistic that separates average exposure from vertical and horizontal sorting, and quantify it in a small-open-economy HANK model with non-homothetic consumption baskets and imported inputs. Vertical sorting amplifies the short-run contraction because low-expenditure households are both more exposed and less able to smooth the shock, while horizontal basket heterogeneity matters mainly for incidence. Imported-input tariffs drive the fall in domestic activity, while tariffs on final goods generate the unequal cost-of-living burden. In sum, household exposure has a moderate aggregate effect but a large distributional one.

1 Introduction

What is a tariff shock? In macroeconomic models, the standard answer is a cost shock. A tariff raises import prices, increases firms' marginal costs, pushes up inflation, and creates a trade-off between price stability and economic activity. With imported intermediate inputs, this logic is immediate: the tariff enters production costs and shifts the Phillips curve upward (Werning et al., 2025). In richer production networks, foreign price shocks propagate through input-output linkages, trade imbalances, and sectoral exposure to foreign inputs (Aguilar et al., 2026).

This view is correct, but incomplete. A tariff is also a household shock. When the price of imported consumption goods rises, a given nominal income buys less than before. Before wages adjust, before households substitute, and before monetary and fiscal policy respond, there is a direct loss of purchasing power.

That loss is not distributed uniformly. Households consume different baskets, so the same tariff produces different cost-of-living increases. They also differ in their ability to smooth shocks: A

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liquid household can absorb a temporary price increase by saving less, while a household close to its borrowing constraint must cut expenditure immediately. The aggregate demand effect of a tariff therefore depends not only on average import exposure, but on who is exposed.

This paper studies that interaction. The central object is the joint distribution of tariff exposure and household spending responses. In other words, a tariff is more contractionary when the households facing the largest purchasing-power losses are also those with the largest intertemporal marginal propensities to consume.

I first formalize this idea in partial equilibrium, reaching an aggregate demand sufficient statistic. The statistic can be decomposed into average exposure, vertical sorting across observable household groups, and horizontal sorting within those groups. This decomposition separates unequal incidence from aggregate amplification: Dispersion in exposure is sufficient for households to bear different tariff burdens, but it is not sufficient for a larger aggregate expenditure response. Amplification requires exposure to be systematically concentrated among households with stronger spending responses.

I then embed the mechanism in a small-open-economy HANK model. To identify the role of exposure heterogeneity, I compare three economies. In the *common-basket economy*, every household has the same import expenditure share. In the *vertical-exposure economy*, import shares vary with household expenditure but not within expenditure groups. In the *full-exposure economy*, households differ both across and within expenditure groups. Each economy is solved separately and has its own stationary distribution and household policies.

Hence, the paper makes a triple contribution: providing a sufficient statistic linking tariff exposure and intertemporal spending responses, separating vertical and horizontal exposure sorting and clarifies when each component affects incidence or aggregate demand, and quantifying the mechanism in a heterogeneous-agent open economy in which household exposure, iMPCs, expenditure switching, imported inputs, and external adjustment interact in general equilibrium.

The main results are the following: **First**, average exposure understates the direct household demand effect. In the baseline calibration, sorting between exposure and spending responses makes the impact purchasing-power effect 11.3 percent larger than the average-exposure benchmark. Almost all of this amplification comes from vertical sorting. Low-expenditure households are simultaneously more exposed to imported goods, less liquid, and more responsive to resource losses. Horizontal basket dispersion matters for incidence, but contributes little to aggregate amplification when it is not systematically sorted with household iMPCs.

Second, the distributional effects are much larger than the aggregate differences across economies. The tariff raises the cost of living of the bottom expenditure decile by 2.86 percent, compared with 2.02 percent for the top decile. Composite consumption falls by 9.57 percent for the bottom decile and by 4.23 percent for the top. Exposure and liquidity reinforce each other: the households facing the largest price increase are also the least able to smooth it.

Third, the tariff on imported intermediate inputs accounts for most of the fall in domestic value added. The tariff on final consumption imports accounts for most of the decline in household imports and generates the heterogeneous purchasing-power channel. A tariff should therefore be read as a composite shock: a cost-of-living shock for households and a production-cost shock for firms.

The main conclusion is simple: Who is exposed matters. The distribution of import exposure has only a moderate effect on the size of the aggregate recession, but a large effect on its composition and on who bears the loss. A representative import share misses both the sorting that amplifies short-run domestic contraction and the household-level inequality hidden beneath the aggregate response.

Related literature. This paper connects four strands of literature. The first studies the household incidence of trade shocks through consumption prices. A central insight is that changes in trade prices do not affect all households equally because households buy different baskets. Porto (2006) and Nicita (2009) develop early survey-based approaches that combine consumption-price effects with income and factor-market channels. Fajgelbaum and Khandelwal (2016) show that non-homothetic expenditure patterns make the gains from trade unequal across the income distribution. Cravino and Levchenko (2017) document heterogeneous cost-of-living effects following large devaluations, arising both across broad consumption categories and within categories.

More directly related to trade protection, Artuc et al. (2021) quantify household exposure to agricultural tariffs, while Jaccard (2022) emphasizes the role of household-level substitution in measuring tariff incidence. Borusyak and Jaravel (2024) show that trade shocks can generate substantial horizontal inequality: households with similar income can face very different price changes because they consume different goods.

This paper builds on that consumer-side view, but shifts the object from exposure alone to exposure interacted with spending responses. Unequal exposure determines who bears the direct cost of a tariff. Its aggregate demand effect additionally depends on whether the exposed households have high or low marginal propensities to spend. The paper therefore moves from a distribution of cost-of-living losses to a joint distribution of losses and spending responses.

The second literature studies heterogeneous marginal propensities to consume and aggregate transmission. Empirical work documents large and persistent differences in household spending responses. Parker et al. (2013) estimate sizeable consumption responses to the 2008 U.S. stimulus payments, while Jappelli and Pistaferri (2014) review the broader evidence on MPC heterogeneity. Kaplan and Violante (2014) show that a model with liquid and illiquid assets can generate large responses to temporary transfers. Kaplan et al. (2014) document the importance of wealthy hand-to-mouth households, who own illiquid wealth but hold little liquid wealth and therefore respond strongly to transitory resource changes.

HANK models embed these spending responses in general equilibrium. Auclert (2019) decomposes monetary transmission into intertemporal substitution and redistribution across households with different MPCs, balance sheets, and income exposures. Kaplan et al. (2018) show that changes in labor income, profits, and asset returns can dominate the direct intertemporal-substitution channel. Auclert et al. (2024) develop the intertemporal Keynesian cross, in which matrices of intertemporal MPCs summarize the propagation of household income changes into aggregate demand.

The sufficient statistic in this paper applies the same logic to a different impulse. A tariff raises the expenditure required to purchase a household's consumption basket. The resulting aggregate demand effect depends on how those purchasing-power losses are allocated across the distribution of intertemporal MPCs. The relevant object is not the average MPC or average exposure separately, but their expenditure-weighted interaction.

The third literature asks when household heterogeneity matters for aggregate fluctuations. Krusell and Smith (1998) provide the classic benchmark. In their incomplete-markets growth model, the wealth distribution has large household-level consequences, but aggregate prices can be forecast accurately using little more than aggregate capital. Heterogeneity is central for distributional outcomes while its effect on aggregate dynamics is limited. This approximate-aggregation result established an important distinction: microeconomic heterogeneity does not automatically imply macroeconomic non-aggregation.

The same question reappears in New Keynesian models. The answer, however, is shock-specific: In the calibrated two-asset model of Kaplan et al. (2018), HANK and RANK produce similar responses to some demand and productivity shocks but differ sharply for monetary and fiscal shocks. Even when aggregate impulse responses are close, the underlying mechanisms can differ because HANK consumption responds less through intertemporal substitution and more through household income.

Bilbiie (2025) provides an analytical HANK framework that nests RANK and isolates the wedges generated by cyclical inequality, self-insurance, and cyclical risk. In that framework, heterogeneity amplifies aggregate demand when high-MPC households are more exposed to the contraction. If inequality or risk does not move in the relevant direction, the aggregate economy can instead look close to RANK.

Debortoli and Galí (2025) reach a related conclusion using nested quantitative models. A suitably calibrated TANK model can reproduce many of the aggregate implications of richer HANK economies because much of the relevant heterogeneity can be summarized by the distinction between constrained and unconstrained households. They also show that policy rules which stabilize inflation strongly compress the differences in aggregate output across RANK, TANK, and HANK. This does not make heterogeneity irrelevant for consumption distributions, asset returns, or welfare. It makes its relevance conditional on the aggregate object being studied.

Recent empirical work pushes the aggregation result further. Bilbiie et al. (2025) construct model-consistent sufficient statistics using matched Norwegian data on income, wealth, and actual consumption. They find little aggregate-demand amplification from heterogeneity once exposure is measured using disposable income after taxes and transfers. High-MPC households have more cyclical labor earnings, but the tax-and-transfer system insures much of that differential exposure. Their result illustrates why heterogeneity must be evaluated jointly with the shock and the policy environment: large MPC dispersion alone is not sufficient for aggregate amplification.

The fourth literature studies tariffs and foreign-price shocks in open-economy macroeconomic models. Evidence from recent U.S. tariffs documents substantial pass-through into duty-inclusive import prices and important effects on prices, trade flows, and welfare (Amiti et al., 2020). Erceg et al. (2018) show that the short-run effects of trade policy depend on nominal rigidities, expenditure switching, and the monetary response. Lindé and Pescatori (2019) revisit Lerner symmetry in a quantitative New Keynesian open economy. Barattieri et al. (2021) show that protectionism behaves like an adverse supply shock, raising inflation while reducing economic activity.

Open-economy HANK models add household real-income and multiplier channels to the standard expenditure-switching mechanism. Auclert et al. (2021b) show that exchange-rate shocks operate through expenditure switching, heterogeneous real-income effects, and aggregate demand feedback. Their use of non-homothetic consumption baskets is particularly relevant here: international price changes have stronger real-income effects when imported goods represent a larger share of the expenditure of high-MPC households.

Recent tariff models emphasize the distinction between final and intermediate imports. Tariffs on consumption goods raise household living costs and induce expenditure switching. Tariffs on imported inputs raise production costs and reduce domestic activity. Auclert et al. (2025) study the short-run effects of tariffs on GDP and the trade balance in an open economy with intermediate-input trade, while Auray et al. (2025) emphasize how nominal rigidities and the monetary regime shape the macroeconomic response to trade conflict. This paper adds a distributional household block to that composite-shock view.

Roadmap. The rest of the paper is organized as follows: Section 2 isolates the direct purchasing-power channel and derives a sufficient statistic for the partial-equilibrium demand effect of a tariff. Section 3 embeds the mechanism in a heterogeneous-agent small open economy. Section 4 studies the quantitative version of the model. Section 5 describes the main findings. Section 6 concludes.

2 What is a tariff for a household?

Locally, a real resource loss. To show this, we isolate the direct fixed-basket cost-of-living channel of tariffs. Through this section, initial expenditure baskets are predetermined[1], non-tariff prices and incomes are held fixed, and there is no contemporaneous fiscal recycling. Therefore, the object is the local purchasing-power loss generated by a tariff-induced increase in household cost-of-living indices, and how this loss maps into current aggregate demand.

Figure 1 places this object within the broader anatomy of a tariff shock. Tariffs can trigger many adjustment margins, but the partial-equilibrium exercise focuses on the purchasing-power/demand branch: prices rise, households are exposed through their consumption baskets, cost of living increases, and spending responds. The key question is whether the households most exposed to the shock are also those with stronger spending responses.

[1]This assumption does not imply that the household consumes the same basket across all periods.

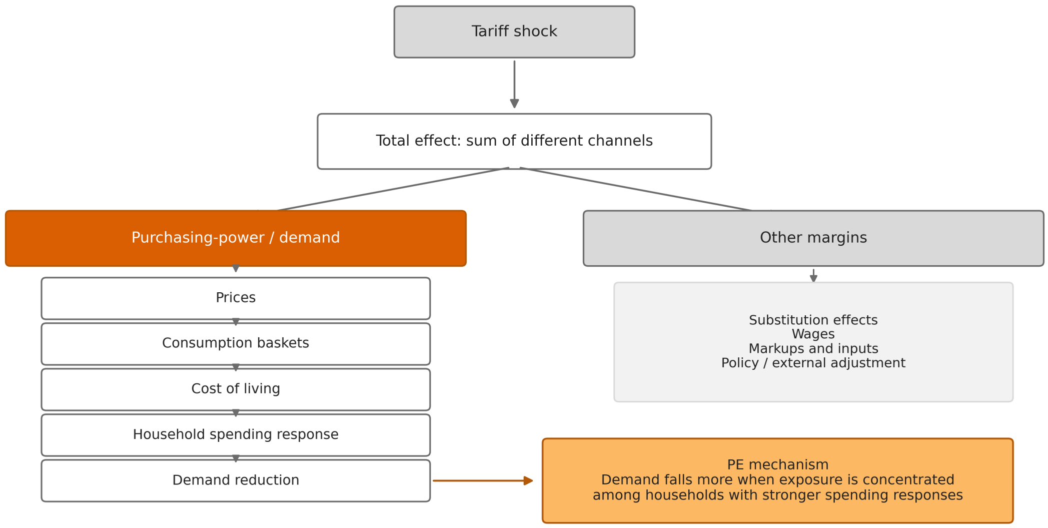

Figure 1: Anatomy of a tariff shock.

Household exposure. Consider a continuum of households indexed by $i \in \mathcal{I}$, with distribution μ. Household i has initial real consumption $c_i > 0$, and aggregate consumption is

$$C = \int_{\mathcal{I}} c_i \, d\mu(i). \tag{1}$$

Households consume goods indexed by $g \in \mathcal{G}$. Let ω_{ig} denote household i's initial expenditure share on good g, with $\sum_g \omega_{ig} = 1$. A local tariff innovation $d\tau_0$ induces the consumer-price path

$$d \log p_{g,t} = \theta_g q_t d\tau_0, \tag{2}$$

where q_t captures the persistence of the tariff-induced price change and θ_g measures the exposure of good g to the tariff. Under full pass-through on tariffed imported goods[2], θ_g is the relevant tariffed import intensity. More generally, it is the first-order consumer-price exposure of good g.

Household i's tariff exposure is the basket-weighted exposure

$$\alpha_i^\tau = \sum_{g \in \mathcal{G}} \omega_{ig} \theta_g. \tag{3}$$

Thus the local change in household i's cost-of-living index is

$$d \log P_{i,t} = \alpha_i^\tau q_t d\tau_0. \tag{4}$$

The exposure α_i^τ is tariff-specific. It depends on the goods affected by the tariff, pass-through into consumer prices, and the household's initial basket.

[2]This assumption simplifies the algebra, but is not necessary.

Assuming that $P_{i,ss} = 1$, the first-order increase in the cost of household i's fixed expenditure base at date s is

$$d(P_{i,s}c_i) = \alpha_i^\tau c_i q_s d\tau_0. \tag{5}$$

For the direct purchasing-power channel, the tariff path is therefore equivalent to a sequence of household-specific resource losses,

$$dy_{i,s}^{eq} = -\alpha_i^\tau c_i q_s d\tau_0. \tag{6}$$

This equivalence is local and fixed-basket. It does not imply that tariffs are globally equivalent to income shocks. It only characterizes the direct cost-of-living component highlighted in Figure 1: for a given initial expenditure base, households with larger tariff exposure experience a larger loss of purchasing power.

Demand incidence. The purchasing-power loss matters for aggregate demand only to the extent that households transmit it into current spending. Let

$$M_{i,0s} = \left.\frac{\partial c_{i,0}}{\partial y_{i,s}}\right|_{ss} \tag{7}$$

denote household i's date-0 consumption response to a one-unit liquid-resource transfer at date s. For the tariff path $q = \{q_s\}_{s\geq 0}$, define the tariff-relevant intertemporal marginal propensity to consume, or iMPC, as

$$\Psi_i(q) = \sum_{s=0}^{\infty} M_{i,0s} q_s. \tag{8}$$

The date-0 consumption response of household i to the tariff-induced purchasing-power losses is

$$dc_{i,0}^{PP} = \sum_{s=0}^{\infty} M_{i,0s} dy_{i,s}^{eq} = -\Psi_i(q)\alpha_i^\tau c_i d\tau_0. \tag{9}$$

Define the consumption-weighted expectation

$$E^C[x_i] = \int_{\mathcal{I}} x_i \frac{c_i}{C}\, d\mu(i). \tag{10}$$

Aggregating across households gives the direct purchasing-power component of the aggregate consumption response:

$$\frac{dC_0^{PP}}{C} = -s_\tau(q)d\tau_0, \qquad s_\tau(q) = E^C[\Psi_i(q)\alpha_i^\tau]. \tag{11}$$

Proposition 1 (Demand sufficient statistic)**.** *For a local tariff innovation $d\tau_0$, holding fixed initial baskets and abstracting from general-equilibrium feedbacks, the direct aggregate demand effect is governed by*

$$s_\tau(q) = E^C[\Psi_i(q)\alpha_i^\tau]. \tag{12}$$

Proof. The tariff-induced loss at date s is $dy_{i,s}^{eq} = -\alpha_i^\tau c_i q_s d\tau_0$. Multiplying by the household's date-0 response $M_{i,0s}$ and summing over s gives $dc_{i,0}^{PP} = -\Psi_i(q)\alpha_i^\tau c_i d\tau_0$. Integrating across households and dividing by C yields the result. $\square$

The statistic separates into an average-exposure benchmark and a sorting term. This is the main mechanism behind the demand branch in Figure 1: the fall in demand is larger when tariff exposure is concentrated among households with stronger spending responses. Let

$$\bar{\Psi}(q) = E^C[\Psi_i(q)], \qquad \bar{\alpha}^\tau = E^C[\alpha_i^\tau]. \tag{13}$$

Then

$$s_\tau(q) = \bar{\Psi}(q)\bar{\alpha}^\tau + Cov^C(\Psi_i(q), \alpha_i^\tau). \tag{14}$$

Corollary 1 (Demand orthogonality)**.** *Average tariff exposure is sufficient for the direct aggregate demand response, given $\bar{\Psi}(q)$, if and only if*

$$Cov^C(\Psi_i(q), \alpha_i^\tau) = 0. \tag{15}$$

If the covariance is positive, the average-exposure benchmark understates the fall in current aggregate consumption. If the covariance is negative, the benchmark overstates it.

Vertical and horizontal incidence. The covariance term can arise across observable groups or within them. This distinction matters because empirical incidence exercises often observe average exposure by income, expenditure, liquidity, or regional groups, but not the identity of the exposed households within each group. One view emphasizes vertical incidence: low- and high-income households buy different bundles, so changes in trade prices affect their cost of living differently (Fajgelbaum and Khandelwal, 2016; Cravino and Levchenko, 2017). A complementary view emphasizes horizontal incidence: households with similar income can still have very different exposure to trade shocks because they consume different goods (Borusyak and Jaravel, 2024). The decomposition below maps this distinction into the tariff-demand statistic.

Let $b(i) \in \mathcal{B}$ denote an observable partition of households, such as income bins, expenditure bins, liquidity groups, or regions. For any household-level variable x_i, define

$$E_b^C[x_i] = E^C[x_i \mid b(i) = b], \tag{16}$$

and let

$$\bar{\Psi}_b(q) = E_b^C[\Psi_i(q)], \qquad \bar{\alpha}_b^\tau = E_b^C[\alpha_i^\tau]. \tag{17}$$

The first object is the average tariff-relevant spending response of group b; the second is its average tariff exposure.

By the law of total covariance,

$$Cov^C(\Psi_i(q), \alpha_i^\tau) = Cov^C\left(\bar{\Psi}_{b(i)}(q), \bar{\alpha}_{b(i)}^\tau\right) + E^C\left[Cov_b^C(\Psi_i(q), \alpha_i^\tau)\right]. \tag{18}$$

Combining this identity with (14) gives

$$s_\tau(q) = \bar{\Psi}(q)\bar{\alpha}^\tau + V_\tau(q) + H_\tau(q), \tag{19}$$

where

$$V_\tau(q) = Cov^C\left(\bar{\Psi}_{b(i)}(q), \bar{\alpha}^\tau_{b(i)}\right) \tag{20}$$

is vertical sorting, and

$$H_\tau(q) = E^C\left[Cov^C_b(\Psi_i(q), \alpha^\tau_i)\right] \tag{21}$$

is horizontal sorting.

Vertical sorting captures whether groups with larger average spending responses also have larger average exposure to tariffed goods. Horizontal sorting captures whether the same relationship holds within observable groups. Thus, $H_\tau(q) > 0$ means that, conditional on belonging to the same group, more exposed households are also the households that cut current spending more strongly after the tariff-induced loss of purchasing power.

Proposition 2 (Vertical and horizontal tariff incidence)**.** *For a local tariff innovation $d\tau_0$, holding fixed initial baskets and abstracting from general-equilibrium feedbacks,*

$$\frac{dC_0^{PP}}{C} = -\left[\bar{\Psi}(q)\bar{\alpha}^\tau + V_\tau(q) + H_\tau(q)\right] d\tau_0. \tag{22}$$

Aggregate average exposure is sufficient, given $\bar{\Psi}(q)$, if and only if

$$V_\tau(q) + H_\tau(q) = 0. \tag{23}$$

Group-level averages are sufficient if and only if

$$H_\tau(q) = 0. \tag{24}$$

The proof follows directly from Proposition 1 and the law of total covariance.

Corollary 2 (Group data may not be enough)**.** *Suppose two economies have the same aggregate exposure, the same average exposure in every observable group, and the same average spending response in every observable group. If they differ in $H_\tau(q)$, they have different direct aggregate demand responses to the same tariff shock.*

The corollary clarifies the empirical content of the decomposition. Group-level income gradients identify the vertical component, but they do not discipline the horizontal component. If $H_\tau(q) \neq 0$, two incidence exercises based on the same group averages can imply different aggregate demand effects because the identity of the exposed households inside each group matters.

A useful special case makes the point transparent. Suppose average tariff exposure is the same across observable groups:

$$\bar{\alpha}^\tau_b = \bar{\alpha}^\tau \qquad \text{for all } b. \tag{25}$$

Then

$$V_\tau(q) = 0, \tag{26}$$

but

$$s_\tau(q) = \bar{\Psi}(q)\bar{\alpha}^\tau + H_\tau(q). \tag{27}$$

Thus, even with no exposure gradient across observable groups, tariff-demand incidence can differ from the average-exposure benchmark. What matters is not only which groups are exposed, but also which households within those groups are exposed.

3 The HANK model

Does the household-incidence mechanism survive in general equilibrium? In this section, we embed the mechanism in a small-open-economy HANK model. The model has one non-standard ingredient: households differ in their exposure to imported consumption goods. Exposure varies systematically with household expenditure and, in the full model, also differs across households with similar expenditure. Specifically, the first component generates vertical exposure heterogeneity; while the second generates horizontal exposure heterogeneity.

The aggregate structure follows the open-economy HANK framework of Auclert et al. (2021b) and the quantitative small-open-economy model of Druedahl et al. (2024).

The model, whose detailed explanation could be found in Appendix A, is designed to answer a distributional question: How the joint distribution of import exposure and spending propensities changes household incidence, aggregate expenditure, and the general-equilibrium response to a tariff?

Households and permanent heterogeneity. Households are indexed by liquid assets a, idiosyncratic labor productivity e, a permanent discount-factor type β, and a permanent consumption-basket type ξ. The individual state is

$$z_{i,t} = (a_{i,t-1}, e_{i,t}, \beta_i, \xi_i)\,. \tag{28}$$

where labor productivity follows a finite-state Markov chain with transition matrix Π_e.

Although not strictly necessary for the proposed mechanism, heterogeneity in β helps the model reproduce the observed dispersion in liquid wealth and marginal propensities to consume. The basket type ξ serves a different purpose. It shifts the household's demand for imported goods without directly changing earnings risk, patience, or access to financial markets.

In the baseline, β_i and ξ_i are independent ex ante. This implies that the model does not impose that households with high MPCs also have import-intensive tastes. Any relationship between exposure and MPCs must arise through household resources and equilibrium sorting. Positive and negative dependence between patience and basket type are considered as robustness exercises.

Households supply a common number of hours N_t, while individual labor income is proportional to productivity $e_{i,t}$. A labor union chooses aggregate hours and sets wages subject to nominal rigidity. This keeps the household block as a tractable one-asset consumption–saving problem while preserving an aggregate labor-supply margin.

All real quantities are expressed in units of the domestic final good. Let $p^C_{M,t}$ denote the tariff-inclusive relative price of imported consumption goods. Household i solves

$$V_t(a_-, e; \beta, \xi) = \max_{c,a'} u(c) - v(N_t) + \beta\, \mathbb{E}_t V_{t+1}(a', e'; \beta, \xi), \tag{29}$$

$$\text{s.t.} \qquad \zeta_t + \mathcal{P}_{\xi,t} c + a' = (1 + r^a_t) a_- + (1 - \tau^\ell_t) w_t e N_t + T_{i,t}, \qquad a' \geq \underline{a}. \tag{30}$$

Here r^a_t is the realized return on the household's liquid claim, w_t is the real wage, τ^ℓ_t is the labor-income tax rate, and $T_{i,t}$ denotes transfers. The terms ζ_t and $\mathcal{P}_{\xi,t}$ are the subsistence expenditure requirement and the price of the supernumerary consumption composite. They are derived below.

Incomplete markets generate heterogeneous spending responses through four margins: liquid wealth, earnings risk, the borrowing constraint, and permanent patience. The tariff interacts with these endogenous spending propensities through heterogeneous consumption baskets.

Non-homothetic consumption baskets. A homothetic CES aggregator implies that, at given relative prices, all households with the same preference weight allocate the same expenditure share to imports. Such a model can generate heterogeneous MPCs, but not a systematic relationship between household resources and import exposure. Every household faces the same proportional purchasing-power loss.

That restriction removes the mechanism studied in this paper. We instead use a Stone–Geary CES aggregator. Household i consumes domestic and imported final goods, denoted by $c_{H,i,t}$ and $c_{M,i,t}$:

$$c_{i,t} = \left[(1 - \omega_{\xi_i})^{1/\eta_C} (c_{H,i,t} - \bar{c}_H)^{\frac{\eta_C - 1}{\eta_C}} + \omega_{\xi_i}^{1/\eta_C} (c_{M,i,t} - \bar{c}_M)^{\frac{\eta_C - 1}{\eta_C}} \right]^{\frac{\eta_C}{\eta_C - 1}}. \tag{31}$$

The parameter η_C is the elasticity of substitution between domestic and imported final goods. The quantities $\bar{c}_H$ and $\bar{c}_M$ are common subsistence requirements. The parameter ω_{ξ_i} is the supernumerary import weight associated with basket type ξ_i.

The common subsistence component generates an expenditure gradient in import exposure. Variation in ω_ξ generates residual basket dispersion among households with similar expenditure. The two sources of exposure heterogeneity are therefore distinct by construction.

The expenditure function is

$$x_{i,t} = \zeta_t + \mathcal{P}_{\xi_i,t} c_{i,t}, \tag{32}$$

where

$$\zeta_t = \bar{c}_H + p^C_{M,t} \bar{c}_M \tag{33}$$

and

$$\mathcal{P}_{\xi,t} = \left[(1 - \omega_\xi) + \omega_\xi (p^C_{M,t})^{1-\eta_C} \right]^{\frac{1}{1-\eta_C}}. \tag{34}$$

The resulting import expenditure share is

$$s_{M,i,t} = \frac{p^C_{M,t}\bar{c}_M + \vartheta_{\xi_i,t}\mathcal{P}_{\xi_i,t}c_{i,t}}{\zeta_t + \mathcal{P}_{\xi_i,t}c_{i,t}}, \tag{35}$$

with

$$\vartheta_{\xi,t} = \omega_\xi \left(\frac{p^C_{M,t}}{\mathcal{P}_{\xi,t}}\right)^{1-\eta_C}. \tag{36}$$

At fixed prices and basket type,

$$\frac{\partial s_{M,i,t}}{\partial \log x_{i,t}} = \vartheta_{\xi_i,t} - s_{M,i,t}. \tag{37}$$

When imported goods are necessity-like, $s_{M,i,t} > \vartheta_{\xi_i,t}$, the import share declines with household expenditure. Low-expenditure households are therefore more exposed to a rise in import prices.

This mechanism aligns exposure with spending responses. Low-expenditure households tend to hold fewer liquid assets, are more likely to be close to the borrowing constraint, and have higher MPCs. Stone–Geary demand can therefore generate a positive exposure–MPC covariance even when basket type and patience are independent.

Horizontal basket dispersion has a different implication. Variation in ω_ξ creates unequal exposure among households with similar resources. This necessarily affects incidence. It affects aggregate demand only when the residual exposure differences are systematically related to household MPCs.

Three economies. The quantitative analysis compares three economies. In the *common-basket economy*, all households have the same import expenditure share in the stationary equilibrium. This economy preserves incomplete markets and MPC heterogeneity but removes household exposure heterogeneity.

In the *vertical-exposure economy*, the Stone–Geary expenditure gradient is active, but all households share the same supernumerary import weight. Import exposure varies with household expenditure, while households with similar expenditure have the same basket.

In the *full-exposure economy*, both the Stone–Geary gradient and permanent basket types are active. Exposure varies across expenditure groups and within them.

The aggregate import share and all non-exposure parameters are held fixed across the three economies. Each economy has its own stationary household distribution, policy functions, and sequence-space Jacobians. The comparisons are therefore equilibrium counterfactuals, not ex post reweighting exercises. The difference between the common-basket and vertical-exposure economies identifies the contribution of the expenditure gradient. The difference between the vertical and full economies identifies the additional contribution of horizontal basket dispersion.

Tariffs and household exposure. Let

$$q_t \equiv \frac{E_t P^*_t}{P_{H,t}} \tag{38}$$

denote the foreign price level in units of the domestic final good. Under producer-currency pricing and full pass-through, the tariff-inclusive prices of imported consumption goods and imported inputs are

$$p^C_{M,t} = (1+\tau^C_t)q_t, \qquad p^I_{M,t} = (1+\tau^I_t)q_t. \tag{39}$$

The distinction between τ^C_t and τ^I_t separates two channels: A tariff on final imports raises household living costs, while a tariff on imported inputs raises production costs. Similarly, a common tariff activates both channels.

Imports are valued externally at the border price q_t. The tariff-inclusive prices determine the cost paid by domestic households and firms, but the tariff wedge is domestic fiscal revenue rather than a payment to foreign suppliers.

By Shephard's lemma, the local expenditure required to compensate household i for a change in the consumption-import tariff satisfies

$$\alpha^\tau_{i,t} \equiv \frac{\partial \log x_{i,t}}{\partial \tau^C_t} = \kappa_{\tau,t} s_{M,i,t}, \tag{40}$$

where

$$\kappa_{\tau,t} \equiv \frac{\partial \log p^C_{M,t}}{\partial \tau^C_t}. \tag{41}$$

The household import share is therefore the model counterpart of the exposure measure introduced in Section 2.

Exposure is endogenous in the HANK model. It depends on household expenditure, permanent basket type, and equilibrium relative prices. The distribution of spending propensities is also endogenous: it depends on assets, productivity, patience, and the borrowing constraint.

The partial-equilibrium statistic inside HANK. Let $M_{z,0s}$ denote the date-0 expenditure response of a household in stationary state z to one unit of liquid resources received at date s. Let q^τ_s denote the import-price path generated by a unit tariff innovation. Define the tariff-relevant iMPC as

$$\Psi_z(q^\tau) = \sum_{s\geq 0} M_{z,0s} q^\tau_s. \tag{42}$$

Holding household distributions, labor income, returns, transfers, and all non-import prices fixed, the direct purchasing-power component of aggregate expenditure satisfies

$$\frac{dX^{PP}_0}{\bar{X}} = -\mathbb{E}^X\left[\Psi_z(q^\tau)\alpha^\tau_z\right] d\tau^C_0. \tag{43}$$

The proof is reported in Appendix A.4. Equation (43) nests the sufficient statistic derived above. The HANK model supplies the endogenous joint distribution of exposure and iMPCs.

For an observable partition $b(z)$, the statistic can be written as

$$\mathbb{E}^X\left[\Psi_z \alpha^\tau_z\right] = \bar{\Psi}\,\bar{\alpha}^\tau + V_\tau + H_\tau, \tag{44}$$

where

$$V_\tau = \mathrm{Cov}^X\left(\mathbb{E}^X[\Psi_z \mid b(z)], \mathbb{E}^X[\alpha_z^\tau \mid b(z)]\right), \tag{45}$$

$$H_\tau = \mathbb{E}^X\left[\mathrm{Cov}^X\left(\Psi_z, \alpha_z^\tau \mid b(z)\right)\right]. \tag{46}$$

In the quantitative implementation, the baseline partition is defined by household expenditure groups.

The decomposition separates unequal incidence from aggregate amplification. Exposure dispersion is sufficient for households to bear different tariff burdens. It is not sufficient for a larger aggregate expenditure response: Amplification requires exposure to be concentrated among households with stronger spending responses.

General equilibrium adds the responses of wages, employment, taxes, transfers, asset returns, the exchange rate, domestic demand, and the household distribution. The comparison between equation (43) and the full equilibrium response shows how much of the direct household mechanism survives after these margins adjust.

Production. The production block distinguishes gross output from domestic value added. Domestic producers use labor to create value added:

$$X_t = Z_t N_t, \tag{47}$$

where X_t is real GDP and Z_t is aggregate productivity.

Competitive final-good firms combine domestic value added $X_{H,t}$ and imported intermediate inputs $M_{I,t}$:

$$Y_t = \left[(1-\alpha_I)^{1/\nu_I} X_{H,t}^{\frac{\nu_I - 1}{\nu_I}} + \alpha_I^{1/\nu_I} M_{I,t}^{\frac{\nu_I - 1}{\nu_I}}\right]^{\frac{\nu_I}{\nu_I - 1}}. \tag{48}$$

The parameter ν_I governs substitution between domestic value added and imported inputs.

Imported inputs enter gross production but not domestic value added. They therefore affect firms' costs and gross output without entering GDP directly. A tariff on imported inputs raises $p_{M,t}^I$, induces substitution toward domestic inputs, and changes labor demand according to ν_I.

Domestic producers charge a flexible-price markup μ_p over marginal cost. Profits net of the fixed operating cost F accrue to the domestic mutual fund. The fixed cost is part of the stationary production structure and allows the model to jointly reconcile markups, dividends, and household financial wealth.

The domestic final good is used for household consumption, government purchases, and exports. Foreign demand is

$$EX_t = \overline{EX}\, q_t^{\eta_X}, \tag{49}$$

where η_X is the export-price elasticity. Foreign absorption and foreign-currency prices are exogenous.

Wage setting and monetary policy. Nominal rigidity operates through staggered wage setting. A labor union allocates common hours across households and resets nominal wages with a Calvo probability. The wage Phillips curve can be written as

$$\pi_t^W = \bar{\beta}\,\mathbb{E}_t \pi_{t+1}^W + \kappa_W \widehat{\mu}_t^W, \tag{50}$$

where $\widehat{\mu}_t^W$ is the wage-markup gap.

The relevant marginal utility in the wage-setting condition is productivity weighted. An additional hour generates different labor income across productivity states and therefore has different consumption value across households. The full nonlinear wage-setting block is reported in Appendix A.

Product prices are fully flexible in the baseline. This choice keeps the nominal block parsimonious and places the focus on household real-income transmission rather than on optimal inflation stabilization.

Aggregate CPI inflation is constructed from the compensated household expenditure indices. The index therefore incorporates heterogeneous stationary baskets rather than imposing a common representative-household basket.

Monetary policy follows an inertial Taylor rule:

$$i_t = \rho_i i_{t-1} + (1-\rho_i)\left[\bar{i} + \phi_\pi \pi_t^{CPI} + \phi_X \log\left(\frac{X_t}{\bar{X}}\right)\right]. \tag{51}$$

Mutual fund and international assets. Households hold a single liquid claim issued by a competitive mutual fund. The fund invests in domestic equity J_t, government debt B_t, and net foreign assets NFA_t:

$$A_t = J_t + B_t + NFA_t. \tag{52}$$

The fund structure separates household saving from portfolio composition while preserving dividends, valuation effects, public debt, and international asset accumulation.

Domestic equity equals the present value of future dividends. International asset pricing implies a real uncovered-interest-parity condition linking the domestic real return, the foreign return, and the expected evolution of q_t. The baseline does not rely on a debt-elastic risk premium. Stationarity is selected through fiscal adjustment and the terminal conditions of the transition problem.

Net foreign assets evolve with the trade balance. Consumption imports and imported inputs are valued at the border price q_t, not at their tariff-inclusive domestic prices.

Fiscal policy. Tariff revenue is

$$\mathcal{R}_t^\tau = \tau_t^C q_t C_{M,t} + \tau_t^I q_t M_{I,t}. \tag{53}$$

In the baseline, tariff revenue initially reduces public debt. A gradual labor-income tax rule subsequently returns debt to its stationary level.

This closure avoids imposing an immediate transfer that mechanically offsets the household purchasing-power loss. It also avoids treating tariff revenue as a payment to foreign producers. Alternative fiscal closures, including revenue waste and uniform household rebates, are considered in the robustness analysis.

Equilibrium. An equilibrium[3] is a sequence of household policies and distributions, prices and wages, production and trade quantities, policy variables, and financial positions such that

1. Households choose consumption and saving optimally, taking wages, prices, taxes, transfers, and asset returns as given;
2. Firms minimize costs and maximize profits, while the labor union sets wages subject to nominal rigidity;
3. Monetary and fiscal policy satisfy their respective rules and intertemporal constraints;
4. The mutual fund satisfies its balance sheet and the relevant asset-pricing conditions;
5. Domestic-goods, labor, asset, government, and external markets clear.

Solution and quantitative experiments. The stationary household problem is solved using the well-known endogenous-grid method (EGM), from the value-function iteration family. Aggregate transition dynamics are solved with sequence-space Jacobians following Auclert et al. (2021a).

We compute separate stationary equilibria and household Jacobians for the common-basket, vertical-exposure, and full-exposure economies.

The benchmark model allows households to reallocate expenditure immediately between domestic and imported final goods according to the Stone–Geary CES demand system. Robustness exercises vary the substitution elasticity, tariff persistence, imported-input intensity, nominal rigidity, fiscal closure, and the sorting between basket types and household spending propensities.

4 Quantitative setup

The model is quarterly and stationary domestic value added is normalized to one. The calibration is designed to discipline the joint distribution of household spending responses and tariff exposure without forcing the aggregate results.

The three economies share the same aggregate calibration and stationary final-import share. Each economy is solved separately and therefore has its own household policies, stationary distribution, and sequence-space Jacobians. Differences across them are equilibrium counterfactuals, not ex post reweighting exercises.

[3] A more detailed derivation of equilibrium could be found in Appendix A.13.

4.1 Calibration

Calibration strategy. The household block is calibrated in two steps. First, three permanent discount-factor types and their population masses are chosen jointly to match household financial wealth and the average quarterly MPC. Second, the Stone–Geary parameters are chosen to match the aggregate final-import share, the dispersion of mean import shares across expenditure deciles, and the residual dispersion within deciles. The remaining parameters follow standard values in the quantitative small-open-economy HANK literature or are assigned directly. The aggregate backbone is close to Auclert et al. (2021b) and Druedahl et al. (2024).

Permanent basket weights satisfy

$$\omega_\xi = \frac{\exp(\omega_0 + \sigma_\xi \xi)}{1 + \exp(\omega_0 + \sigma_\xi \xi)}, \tag{54}$$

where ξ takes five standardized values. The baseline sorting parameter between patience and basket type is zero. The model therefore does not impose that intrinsically impatient households have more import-intensive tastes. Table 1 reports the structural calibration.

Numerical implementation. The idiosyncratic productivity process is approximated by a seven-state Rouwenhorst chain and the household distribution contains three permanent discount-factor types and five permanent basket types.

The common-basket, vertical-exposure, and full-exposure economies are solved independently. In the common and vertical counterfactuals, the common supernumerary import weight is adjusted so that all three economies reproduce the same aggregate final-import expenditure share.

Moments. Table 2 compares the stationary model with the moments used in the calibration. Panel A contains the targeted moments and Panel B reports external validation moments that are not used to choose any parameter.

The model matches the targeted moments closely. The small difference in government consumption arises because the steady-state government budget, dividends, financial wealth, and the fixed operating cost are solved jointly.

The non-targeted moments are less exact. The model produces a four-quarter cumulative MPC of 0.284, below the reference value of 0.51. It also generates slightly less income inequality and a slightly smaller low-liquidity share than their respective benchmarks. The baseline is therefore conservative with respect to persistent household spending responses. None of the central tariff results relies on targeting an unusually large cumulative MPC.

Table 1: Calibration

Parameter	Description	Value	Source / target
Panel A: Preferences and household heterogeneity			
σ	Relative risk aversion	1.000	Standard
φ	Inverse Frisch elasticity	2.000	Standard
$\{\beta_L, \beta_M, \beta_H\}$	Discount-factor types	$\{0.96891,\ 0.98800,\ 0.99375\}$	Wealth and average MPC
$\{\pi^\beta_L, \pi^\beta_M, \pi^\beta_H\}$	Population masses	$\{0.20,\ 0.35,\ 0.45\}$	Wealth and average MPC
ρ_e	Earnings persistence	0.966	Standard HANK calibration
σ_e	Earnings dispersion	0.500	Standard HANK calibration
Panel B: Household consumption baskets			
η_C	Elasticity between domestic and imported goods	1.500	Standard open-economy value
$\bar{c}_H$	Domestic subsistence requirement	0.02336	Exposure gradient
$\bar{c}_M$	Imported subsistence requirement	0.05664	Exposure gradient
ω_0	Mean import taste	-1.15422	Aggregate import share
σ_ξ	Basket-type dispersion	0.14062	Within-decile dispersion
$\chi_{\beta,\xi}$	Patience–basket sorting	0.000	Baseline independence
Panel C: Production and international trade			
μ_p	Gross product markup	1.200	Standard
ν_I	Elasticity between value added and imported inputs	0.500	Auclert et al. (2021b)
η_X	Export-demand elasticity	2.000	Standard SOE calibration
Panel D: Nominal, monetary, fiscal, and external blocks			
κ_W	Wage Phillips-curve slope	0.050	Druedahl et al. (2024)
ρ_i	Interest-rate smoothing	0.900	Standard Taylor rule
ϕ_π	Response to CPI inflation	1.500	Taylor principle
ϕ_X	Response to domestic value added	0.000	Baseline rule
ψ_B	Labor-tax response to debt	0.020	Debt stabilization
ϕ_{NFA}	External risk-premium elasticity	0.000	Perfect capital mobility
Panel E: Tariff process			
ρ_τ	Tariff persistence	0.750	Baseline experiment
$\Delta\tau_0$	Impact tariff innovation	0.100	Baseline experiment

Notes: The model is quarterly. Parameters calibrated to moments refer to the targets reported in Table 2. The fixed operating cost is determined residually and equals 0.167 of stationary GDP.

Table 2: **Moments**

Moment	Data/reference	Model
Panel A: Targeted moments		
Annual real interest rate	0.0200	0.0200
Average quarterly MPC	0.2000	0.2000
Household financial wealth / quarterly GDP	10.0000	10.0000
Government consumption / GDP	0.2000	0.2029
Government debt / quarterly GDP	2.3200	2.3200
Total imports / GDP	0.4200	0.4200
Imported inputs / total imports	0.5500	0.5500
Gross product markup	1.2000	1.2000
Final-import expenditure share	0.3000	0.3000
SD of expenditure-decile mean import shares	0.0315	0.0315
Mean within-decile SD of import shares	0.0250	0.0250
Panel B: Non-targeted moments		
Four-quarter cumulative MPC	0.5100	0.2836
Disposable-flow income Gini	0.3100	0.2751
Share with very low liquid assets	0.2500	0.2158

Notes: Panel A reports calibration targets. The reference values in Panel B are not targeted. Very low liquid wealth is defined as assets below five percent of mean household flow income. Ratios are expressed as fractions rather than percentages.

4.2 Steady-state exposure and spending responses

Figure 2 shows the joint cross-sectional structure that drives the tariff mechanism. The left panel shows a strong expenditure gradient. Final imports account for 38.4 percent of expenditure in the bottom decile and 27.5 percent in the top decile. The common-basket economy removes this gradient and sets every household's exposure to the aggregate share of 30 percent. The vertical and full economies have almost identical decile means because permanent basket types are calibrated to generate dispersion within expenditure groups, not to change the average gradient across them.

The middle panel shows equally strong heterogeneity in spending responses. The quarterly iMPC is close to one in the bottom liquid-asset decile, falls below 0.12 in the third decile, and approaches zero at the top of the distribution. The average quarterly MPC of 0.20 masks a highly concentrated distribution of short-run responses.

The two forms of heterogeneity are positively aligned. As shown in the right panel, average final-import exposure rises from approximately 28 percent in the lowest-iMPC decile to 37 percent in the highest. The expenditure-weighted correlation between exposure and the quarterly iMPC is 0.439.

This sorting is not imposed through basket tastes. It emerges because low-expenditure households tend to hold fewer liquid assets and devote a larger share of expenditure to necessity-like imports. The model therefore generates the positive covariance required by the sufficient statistic before any aggregate tariff experiment is run.

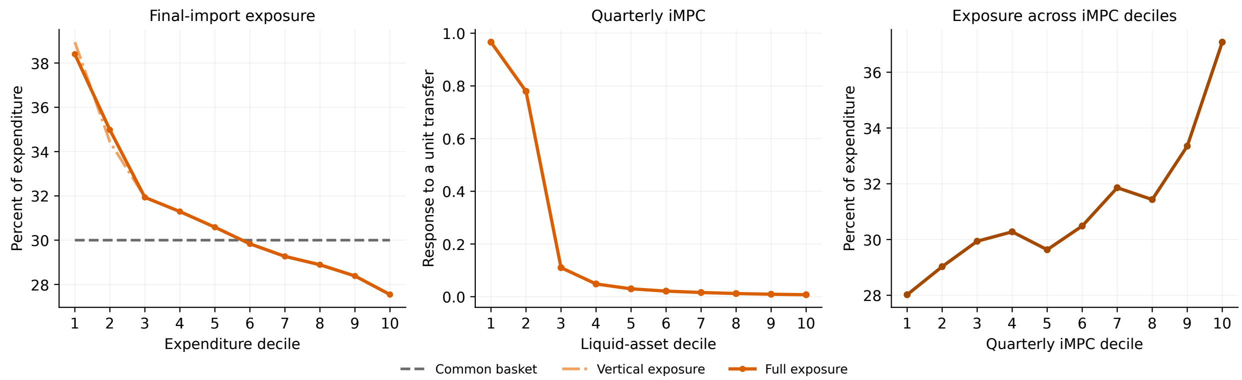

Figure 2: Steady-state household heterogeneity. The left panel reports final-import expenditure shares across expenditure deciles. The middle panel reports quarterly iMPCs across liquid-asset deciles. The right panel reports average final-import exposure across quarterly-iMPC deciles in the full-exposure economy.

5 Results

Aggregate transmission. Figure 3 reports the response to the common tariff shock. As the left hand-side panels show, a tariff shock is contractionary and inflationary, in line with the supply-side effects found in the literature.

In the full-exposure economy, domestic value added falls by 1.56 percent on impact. The responses are front-loaded: Inflation reverses after the initial increase as the tariff wedge fades, domestic demand remains weak, and the relative import price adjusts. Domestic value added begins to recover immediately and temporarily rises above its stationary level. Household consumption and the real wage recover more slowly.

Interestingly, the two tariff margins have sharply different effects. In the full-exposure economy, a tariff applied only to final consumption imports reduces impact output by 0.13 percent. It instead produces a large decline in final-import quantities of 9.54 percent.

A tariff applied only to imported inputs reduces output by 1.43 percent. Imported inputs are difficult to replace because the elasticity between foreign inputs and domestic value added is lower than one. The input tariff therefore acts mainly as a production-cost shock.

To first order, the two experiments add up: The household channel drives the fall in final imports, while the production channel drives the fall in domestic value added.

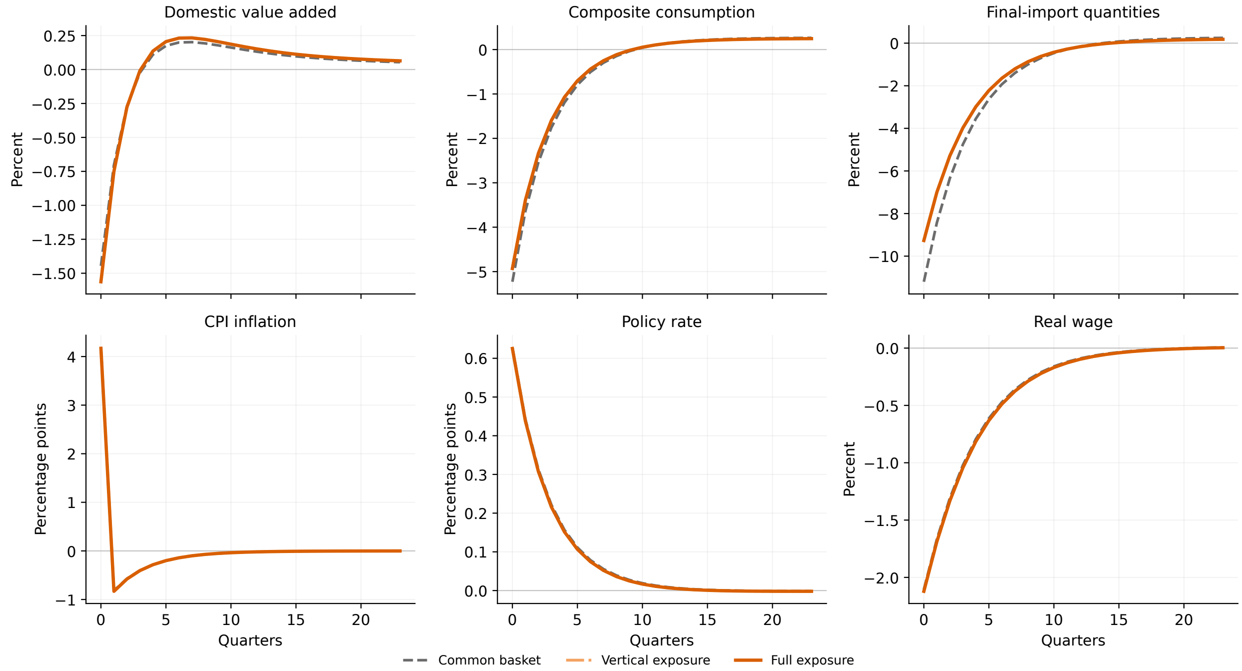

Figure 3: Aggregate responses to a common tariff shock. The figure reports impulse responses to a ten-percentage-point tariff applied jointly to imported final consumption goods and imported intermediate inputs. Real quantities are percentage deviations from stationary values. CPI inflation and the policy rate are expressed in percentage points.

Exposure heterogeneity and the initial contraction. Impact output falls by 1.45 percent in the common-basket economy, 1.56 percent in the vertical-exposure economy, and 1.56 percent in the full-exposure economy. Moving from common exposure to the full distribution therefore makes the initial output response approximately 0.117 percentage points more negative.

Almost the entire difference is already present in the vertical economy. The output responses of the vertical and full economies differ by less than 0.001 percentage points on impact. Residual basket heterogeneity does not create meaningful aggregate amplification in the baseline because it is nearly orthogonal to household iMPCs.

The inflation, interest-rate, and real-wage paths are almost identical across economies. Exposure heterogeneity acts through the composition of household demand rather than through a different aggregate monetary response.

From partial equilibrium to general equilibrium. Figure 4 connects the sufficient statistic to the full equilibrium response.

The average-exposure component reduces household expenditure by 0.382 percent on impact. Vertical sorting contributes an additional 0.040 percent and horizontal sorting adds 0.003 percent. The total direct purchasing-power effect is therefore -0.425 percent. The direct effect is 11.3 percent larger than an average-exposure calculation would imply. The reason is simple: households facing larger cost-of-living increases also have larger spending responses.

The decomposition also reveals a clear hierarchy: More than 90 percent of the amplification

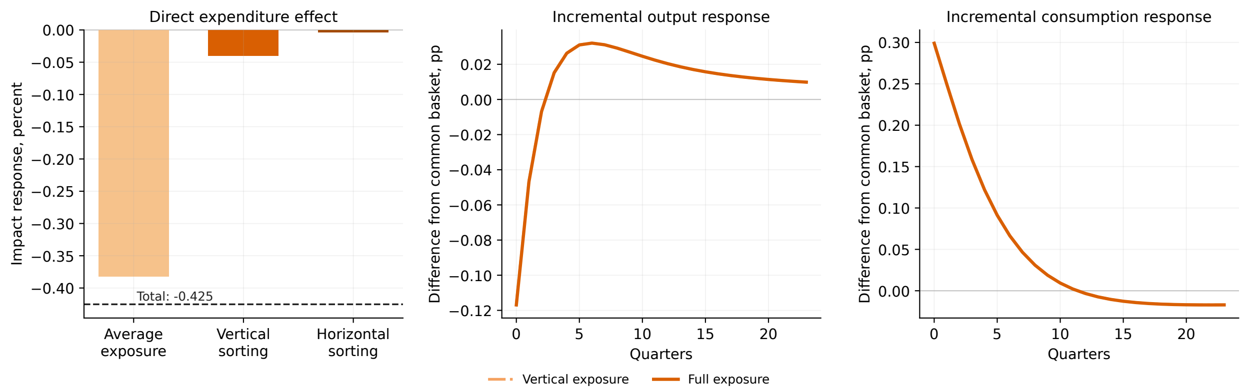

Figure 4: Direct incidence and aggregate amplification. The left panel decomposes the impact direct-expenditure response into average exposure, vertical sorting, and horizontal sorting. The middle and right panels report differences in domestic value added and composite consumption relative to the common-basket economy. Differences are expressed in percentage points.

beyond average exposure comes from vertical sorting. Horizontal exposure dispersion is quantitatively relevant for who bears the tariff, but its aggregate contribution is small when basket type is independent of patience.

Mapping into consumption. The middle panel shows that the household-incidence mechanism survives in domestic value added. On impact, output falls about 0.12 percentage points more in the full-exposure economy than in the common-basket economy. The gap is concentrated in the first few quarters and later changes sign.

The right panel shows that composite consumption behaves differently. Its impact decline is 4.93 percent in the full-exposure economy and 5.23 percent in the common-basket economy. Composite consumption therefore falls approximately 0.30 percentage points less under full exposure.

This is not a contradiction. The sufficient statistic isolates the direct purchasing-power effect while holding income, returns, transfers, and non-import prices fixed. The general-equilibrium response also contains income effects, substitution, monetary tightening, fiscal adjustment, and asset-price movements. The composition of spending is decisive: Final-import quantities fall by 11.20 percent in the common-basket economy but by only 9.27 percent under full exposure. Necessity-like imports make highly exposed households substitute away from imports less aggressively. They preserve more imported consumption but redirect less expenditure toward domestic goods. Domestic consumption therefore falls more, and so does domestic value added, even though the decline in the composite consumption index is smaller.

The macroeconomic contribution of exposure heterogeneity is thus not a uniformly larger response in every aggregate. It is a weaker expenditure-switching response toward domestic production.

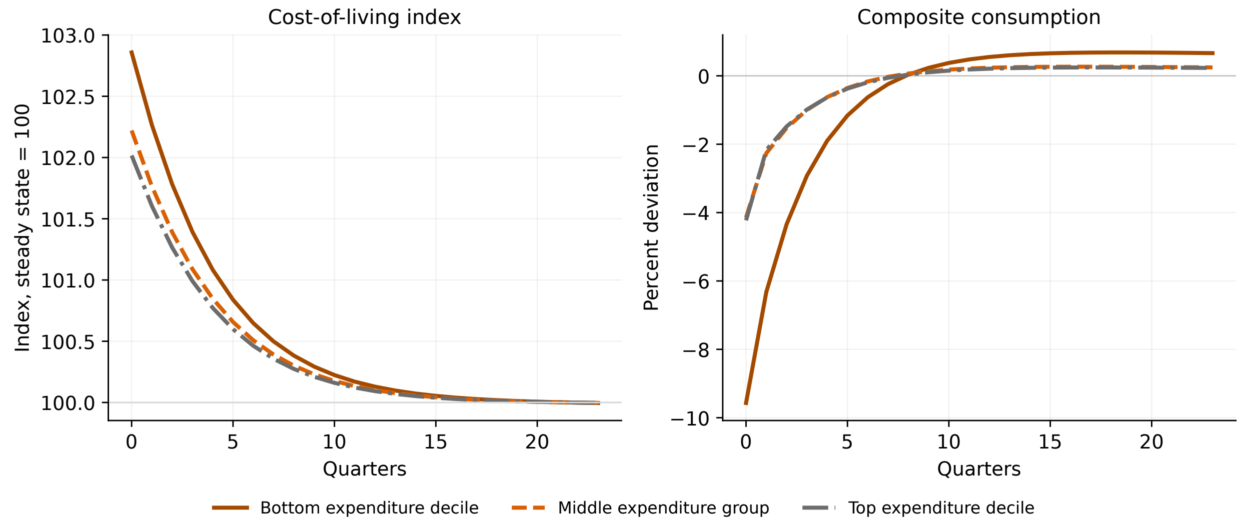

Figure 5: Distributional incidence of the tariff shock. The figure reports responses for the bottom expenditure decile, a middle expenditure group, and the top expenditure decile in the full-exposure economy. The left panel shows compensated cost-of-living indices normalized to 100 in the stationary equilibrium. The right panel reports composite-consumption responses as percentage deviations from stationary values.

5.1 Distributional incidence

The aggregate differences conceal much larger household-level effects. Figure 5 reports the responses of three expenditure groups in the full-exposure economy.

The cost-of-living shock. The compensated cost-of-living index rises by 2.86 percent for the bottom expenditure decile. The corresponding increases are 2.22 percent for the middle group and 2.02 percent for the top decile. These differences arise before household smoothing is considered. The same import-price shock absorbs a larger fraction of the budget of low-expenditure households because imports account for a larger share of their initial consumption baskets.

Exposure and liquidity. The consumption responses are substantially more dispersed than the cost-of-living responses. Composite consumption falls by 9.57 percent for the bottom expenditure decile, compared with 4.15 percent for the middle group and 4.23 percent for the top decile.

The bottom group is hit twice. It faces the largest increase in its cost of living and has the least capacity to smooth the resulting resource loss. Low liquid wealth converts a moderately larger price shock into a much larger consumption response.

The similar impact responses of the middle and top groups also show why exposure alone is not sufficient. The top group has the smallest direct cost-of-living increase, but its consumption remains exposed to indirect changes in wages, asset returns, taxes, and domestic production.

Wrapping up: Incidence is larger than aggregate amplification. The difference between full and common exposure is only 0.117 percentage points for impact output. The difference in consumption losses between the bottom and top expenditure deciles exceeds five percentage points.

Household exposure therefore matters much more for the distribution of the tariff burden than for the size of the aggregate recession. General equilibrium compresses the aggregate differences but does not remove the underlying incidence. The quantitative result is precise. Vertical sorting amplifies the short-run fall in domestic activity. Horizontal basket dispersion mainly changes which households lose. Both dimensions matter, but they matter for different objects.

6 Conclusion

This paper studies how the distribution of household import exposure shapes the macroeconomic effects of tariffs. The central result is that average exposure is not sufficient. The direct expenditure response also depends on whether the households facing the largest cost-of-living losses are those with the strongest spending responses.

I formalize this mechanism through a sufficient statistic that separates average exposure from vertical and horizontal sorting, and quantify it in a small-open-economy HANK model. Vertical sorting amplifies the short-run contraction because low-expenditure households are both more exposed to imported goods and less able to smooth the shock. Horizontal basket heterogeneity matters mainly for the distribution of losses when it is not systematically related to household iMPCs.

The aggregate and distributional implications are therefore different. Imported-input tariffs account for most of the fall in domestic activity, while tariffs on final goods generate the heterogeneous purchasing-power channel. Household exposure has a moderate effect on aggregate output, but a large effect on who bears the tariff. The broader lesson is simple: heterogeneity matters in the aggregate only when exposure and spending responses are aligned, but it can remain first-order for incidence even when aggregate amplification is limited.

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A Full model and derivations

A.1 Prices, tariffs, and timing

Let $P_{H,t}$ denote the price of the domestic final good and let E_t denote the domestic-currency price of foreign currency. The foreign producer price of imports is P_t^*. Define the real exchange rate as

$$q_t \equiv \frac{E_t P_t^*}{P_{H,t}}. \tag{55}$$

An increase in q_t is a real depreciation.

The tariff-inclusive relative prices of imported consumption goods and imported intermediate inputs are

$$p_{M,t}^C = (1+\tau_t^C)q_t, \qquad p_{M,t}^I = (1+\tau_t^I)q_t. \tag{56}$$

The border price is q_t. Domestic households and firms pay the tariff-inclusive prices, but foreign suppliers receive the border price.

Because the steady-state tariff can be zero, tariff shocks are written in terms of the log tariff wedge,

$$\vartheta_t^j \equiv \log(1+\tau_t^j) - \log(1+\bar{\tau}^j), \qquad j \in \{C, I\}. \tag{57}$$

Each wedge follows

$$\vartheta_t^j = \rho_{\tau,j}\vartheta_{t-1}^j + \varepsilon_{\tau,j,t}. \tag{58}$$

The final-goods experiment sets $\varepsilon_{\tau,I,t} = 0$, the input-tariff experiment sets $\varepsilon_{\tau,C,t} = 0$, and the common tariff applies the same innovation to both wedges.

Domestic final-good inflation, nominal wage inflation, and CPI inflation are

$$\Pi_{H,t} \equiv \frac{P_{H,t}}{P_{H,t-1}}, \qquad \Pi_t^W \equiv \frac{W_t}{W_{t-1}}, \qquad \Pi_t^{CPI} \equiv \frac{P_t^{CPI}}{P_{t-1}^{CPI}}. \tag{59}$$

The real wage is

$$w_t \equiv \frac{W_t}{P_{H,t}}. \tag{60}$$

Consequently,

$$\Pi_{H,t} = \Pi_t^W \frac{w_{t-1}}{w_t}. \tag{61}$$

Households enter period t with liquid assets $a_{i,t-1}$. The fund return is realized, labor income and transfers are received, and the household chooses consumption and end-of-period assets $a_{i,t}$.

A.2 Households

Preferences and individual states. Households differ in liquid wealth a, idiosyncratic labor productivity e, a permanent discount factor β, and a permanent basket type ξ. The individual state at the beginning of period t is

$$z_{i,t} = (a_{i,t-1}, e_{i,t}, \beta_i, \xi_i). \tag{62}$$

Labor productivity follows a finite-state Markov chain,

$$\Pr(e_{i,t+1} = e' \mid e_{i,t} = e) = \Pi_e(e, e'). \tag{63}$$

The permanent types β_i and ξ_i do not change over time. Their joint population distribution is denoted by $\pi_{\beta,\xi}$. In the baseline,

$$\pi_{\beta,\xi} = \pi_\beta \pi_\xi. \tag{64}$$

All households work the common number of hours N_t, but their labor income is proportional to productivity $e_{i,t}$. Period utility is

$$U_{i,t} = u(c_{i,t}) - v(N_t), \tag{65}$$

where

$$u(c) = \begin{cases} \dfrac{c^{1-\sigma} - 1}{1-\sigma}, & \sigma \neq 1, \\ \\ \log c, & \sigma = 1, \end{cases} \tag{66}$$

and

$$v(N) = \chi_N \frac{N^{1+\varphi}}{1+\varphi}. \tag{67}$$

Intratemporal consumption problem. Household i consumes the domestic final good $c_{H,i,t}$ and the imported final good $c_{M,i,t}$. The consumption composite is

$$c_{i,t} = \left[(1-\omega_{\xi_i})^{1/\eta_C} (c_{H,i,t} - \bar{c}_H)^{\frac{\eta_C - 1}{\eta_C}} + \omega_{\xi_i}^{1/\eta_C} (c_{M,i,t} - \bar{c}_M)^{\frac{\eta_C - 1}{\eta_C}} \right]^{\frac{\eta_C}{\eta_C - 1}}. \tag{68}$$

The household solves

$$\min_{c_{H,i,t}, c_{M,i,t}} \left\{ c_{H,i,t} + p^C_{M,t} c_{M,i,t} \right\} \tag{69}$$

subject to equation (68) and

$$c_{H,i,t} \geq \bar{c}_H, \qquad c_{M,i,t} \geq \bar{c}_M. \tag{70}$$

Define supernumerary consumption by

$$\widetilde{c}_{H,i,t} \equiv c_{H,i,t} - \bar{c}_H, \qquad \widetilde{c}_{M,i,t} \equiv c_{M,i,t} - \bar{c}_M. \tag{71}$$

The first-order condition for the relative allocation is

$$\frac{\widetilde{c}_{M,i,t}}{\widetilde{c}_{H,i,t}} = \frac{\omega_{\xi_i}}{1-\omega_{\xi_i}} \left(p^C_{M,t}\right)^{-\eta_C}. \tag{72}$$

The supernumerary price index is

$$\mathcal{P}_{\xi,t} = \left[(1-\omega_\xi) + \omega_\xi (p^C_{M,t})^{1-\eta_C} \right]^{\frac{1}{1-\eta_C}}. \tag{73}$$

The cost of the subsistence bundle is

$$\zeta_t = \bar{c}_H + p^C_{M,t} \bar{c}_M. \tag{74}$$

Conditional demands are

$$c_{H,i,t} = \bar{c}_H + (1-\omega_{\xi_i}) \mathcal{P}^{\eta_C}_{\xi_i,t} c_{i,t}, \tag{75}$$

$$c_{M,i,t} = \bar{c}_M + \omega_{\xi_i} \left(\frac{\mathcal{P}_{\xi_i,t}}{p^C_{M,t}} \right)^{\eta_C} c_{i,t}. \tag{76}$$

The minimum expenditure required to obtain composite consumption $c_{i,t}$ is

$$x_{i,t} = \zeta_t + \mathcal{P}_{\xi_i,t} c_{i,t}. \tag{77}$$

Define the supernumerary import expenditure share by

$$\vartheta_{\xi,t} \equiv \omega_\xi \left(\frac{p^C_{M,t}}{\mathcal{P}_{\xi,t}} \right)^{1-\eta_C}. \tag{78}$$

The household's total import expenditure share is

$$s_{M,i,t} = \frac{p^C_{M,t} \bar{c}_M + \vartheta_{\xi_i,t} \mathcal{P}_{\xi_i,t} c_{i,t}}{\zeta_t + \mathcal{P}_{\xi_i,t} c_{i,t}}. \tag{79}$$

Holding prices and basket type fixed,

$$\frac{\partial s_{M,i,t}}{\partial \log x_{i,t}} = \vartheta_{\xi_i,t} - s_{M,i,t}. \tag{80}$$

If imported goods are necessity-like, $s_{M,i,t} > \vartheta_{\xi_i,t}$, the import share falls with household expenditure.

Intertemporal problem. Let r^a_t denote the realized net return on the mutual-fund claim. Given the aggregate state, household i solves

$$V_t(a_-, e; \beta, \xi) = \max_{c,a'} \quad u(c) - v(N_t) + \beta \sum_{e'} \Pi_e(e, e') V_{t+1}(a', e'; \beta, \xi) \tag{81}$$

$$\text{s.t.} \quad \zeta_t + \mathcal{P}_{\xi,t} c + a' = (1 + r^a_t) a_- + (1 - \tau^\ell_t) w_t e N_t + T_t(e, \beta, \xi), \tag{82}$$

$$a' \geq \underline{a}. \tag{83}$$

The Lagrangian is

$$\begin{aligned} L = {} & u(c) - v(N_t) + \beta \sum_{e'} \Pi_e(e, e') V_{t+1}(a', e'; \beta, \xi) \\ & + \lambda \Big[(1 + r^a_t) a_- + (1 - \tau^\ell_t) w_t e N_t + T_t - \zeta_t - \mathcal{P}_{\xi,t} c - a' \Big] \\ & + \mu (a' - \underline{a}). \end{aligned} \tag{84}$$

The first-order condition for composite consumption is

$$u'(c_{i,t}) = \lambda_{i,t} \mathcal{P}_{\xi_i,t}. \tag{85}$$

Hence, the marginal value of one unit of domestic-good resources is

$$\lambda_{i,t} = \frac{u'(c_{i,t})}{\mathcal{P}_{\xi_i,t}}. \tag{86}$$

The first-order condition for end-of-period assets is

$$\lambda_{i,t} = \beta_i \sum_{e'} \Pi_e(e_{i,t}, e') V_{a,t+1}(a_{i,t}, e'; \beta_i, \xi_i) + \mu_{i,t}. \tag{87}$$

The envelope condition is

$$V_{a,t}(a_{i,t-1}, e_{i,t}; \beta_i, \xi_i) = (1 + r_t^a)\lambda_{i,t}. \tag{88}$$

Combining the two conditions gives the Euler inequality

$$\lambda_{i,t} \geq \beta_i \sum_{e'} \Pi_e(e_{i,t}, e')(1 + r_{t+1}^a)\lambda_{i,t+1}(e'), \tag{89}$$

with equality whenever $a_{i,t} > \underline{a}$. The complementary slackness conditions are

$$\mu_{i,t} \geq 0, \qquad a_{i,t} - \underline{a} \geq 0, \qquad \mu_{i,t}(a_{i,t} - \underline{a}) = 0. \tag{90}$$

For CRRA utility, the interior Euler equation implies

$$c_{i,t} = \left[\mathcal{P}_{\xi_i,t} \beta_i \sum_{e'} \Pi_e(e_{i,t}, e')(1 + r_{t+1}^a)\lambda_{i,t+1}(e')\right]^{-1/\sigma}. \tag{91}$$

Given $a_{i,t}$ and the implied $c_{i,t}$, the endogenous current asset position is

$$a_{i,t-1} = \frac{\zeta_t + \mathcal{P}_{\xi_i,t} c_{i,t} + a_{i,t} - (1 - \tau_t^\ell) w_t e_{i,t} N_t - T_t(e_{i,t}, \beta_i, \xi_i)}{1 + r_t^a}. \tag{92}$$

Equations (91)–(92) define the endogenous-grid step used in the numerical solution.

Distribution. Let

$$D_t(a_-, e, \beta, \xi) \tag{93}$$

denote the beginning-of-period distribution. Given the asset policy $a_t' = g_{a,t}(a_-, e, \beta, \xi)$, the distribution evolves according to

$$\begin{aligned} D_{t+1}(a', e', \beta, \xi) = \sum_e \int & \mathbf{1}\left\{g_{a,t}(a_-, e, \beta, \xi) \leq a'\right\} \\ & \times \Pi_e(e, e')\, dD_t(a_-, e, \beta, \xi). \end{aligned} \tag{94}$$

In the discrete implementation, the asset policy is lottery-interpolated onto the fixed asset grid, which preserves total probability mass.

Aggregation. Aggregate end-of-period household assets are

$$A_t = \int g_{a,t}(z)\, dD_t(z). \tag{95}$$

Aggregate composite consumption is

$$C_t = \int c_t(z)\, dD_t(z). \tag{96}$$

Aggregate household expenditure is

$$\mathcal{X}_t = \int x_t(z)\, dD_t(z). \tag{97}$$

Aggregate domestic and imported final consumption are

$$C_{H,t} = \int c_{H,t}(z)\, dD_t(z), \tag{98}$$

$$C_{M,t} = \int c_{M,t}(z)\, dD_t(z). \tag{99}$$

The expenditure identity is

$$\mathcal{X}_t = C_{H,t} + p^C_{M,t} C_{M,t}. \tag{100}$$

Normalize mean labor productivity to one:

$$\int e\, dD_t(z) = 1. \tag{101}$$

Aggregate labor income is therefore

$$w_t N_t. \tag{102}$$

The productivity-weighted marginal value of labor income is

$$\Lambda^e_t = \int e \lambda_t(z)\, dD_t(z). \tag{103}$$

This is the household object entering the wage-setting block.

A.3 Tariff exposure inside the household problem

Differentiate household expenditure with respect to the relative import price while holding composite consumption fixed:

$$\frac{\partial x_{i,t}}{\partial p^C_{M,t}} = \bar{c}_M + \frac{\partial \mathcal{P}_{\xi_i,t}}{\partial p^C_{M,t}} c_{i,t}. \tag{104}$$

The supernumerary price index satisfies

$$\frac{\partial \mathcal{P}_{\xi,t}}{\partial p^C_{M,t}} = \vartheta_{\xi,t} \frac{\mathcal{P}_{\xi,t}}{p^C_{M,t}}. \tag{105}$$

It follows that

$$\frac{\partial \log x_{i,t}}{\partial \log p^C_{M,t}} = s_{M,i,t}. \tag{106}$$

Since

$$p^C_{M,t} = (1 + \tau^C_t) q_t, \tag{107}$$

the tariff exposure of household i is

$$\alpha^\tau_{i,t} \equiv \frac{\partial \log x_{i,t}}{\partial \tau^C_t} = \frac{s_{M,i,t}}{1 + \tau^C_t}. \tag{108}$$

More generally, allowing for incomplete pass-through,

$$\alpha^\tau_{i,t} = \kappa_{\tau,t} s_{M,i,t}, \qquad \kappa_{\tau,t} \equiv \frac{\partial \log p^C_{M,t}}{\partial \tau^C_t}. \tag{109}$$

A.4 The partial-equilibrium statistic inside HANK

Let $\bar{z}$ denote a household state in the stationary distribution. Define

$$M_{\bar{z},0s} \equiv \left.\frac{\partial x_{\bar{z},0}}{\partial y_{\bar{z},s}}\right|_{\text{ss}} \tag{110}$$

as the date-0 expenditure response to one unit of liquid resources received at date s.

Let q^τ_s denote the tariff-induced path of the import-price wedge. The tariff-relevant iMPC is

$$\Psi_{\bar{z}}(q^\tau) = \sum_{s=0}^{\infty} M_{\bar{z},0s} q^\tau_s. \tag{111}$$

Holding the distribution, labor income, transfers, asset returns, and all non-import prices fixed, the tariff generates the resource-equivalent loss

$$dy^{\text{eq}}_{\bar{z},s} = -\alpha^\tau_{\bar{z}} \bar{x}_{\bar{z}} q^\tau_s d\tau^C_0. \tag{112}$$

The date-0 expenditure response of household $\bar{z}$ is

$$\begin{aligned} dx^{PP}_{\bar{z},0} &= \sum_{s=0}^{\infty} M_{\bar{z},0s} dy^{\text{eq}}_{\bar{z},s} \\ &= -\Psi_{\bar{z}}(q^\tau) \alpha^\tau_{\bar{z}} \bar{x}_{\bar{z}} d\tau^C_0. \end{aligned} \tag{113}$$

Define the expenditure-weighted stationary expectation as

$$\mathbb{E}^{\mathcal{X}}[h_{\bar{z}}] = \int h_{\bar{z}} \frac{\bar{x}_{\bar{z}}}{\bar{\mathcal{X}}} \, d\bar{D}(\bar{z}). \tag{114}$$

Aggregating equation (113) gives

$$\frac{d\mathcal{X}_0^{PP}}{\bar{\mathcal{X}}} = -\mathbb{E}^{\mathcal{X}} \left[\Psi_{\bar{z}}(q^\tau) \alpha_{\bar{z}}^\tau \right] d\tau_0^C. \tag{115}$$

For any observable partition $b(\bar{z})$,

$$\mathbb{E}^{\mathcal{X}} \left[\Psi_{\bar{z}} \alpha_{\bar{z}}^\tau \right] = \bar{\Psi} \bar{\alpha}^\tau + V_\tau + H_\tau, \tag{116}$$

where

$$\bar{\Psi} = \mathbb{E}^{\mathcal{X}}[\Psi_{\bar{z}}], \qquad \bar{\alpha}^\tau = \mathbb{E}^{\mathcal{X}}[\alpha_{\bar{z}}^\tau], \tag{117}$$

$$V_\tau = \mathrm{Cov}^{\mathcal{X}} \left(\mathbb{E}^{\mathcal{X}}[\Psi_{\bar{z}} \mid b(\bar{z})], \mathbb{E}^{\mathcal{X}}[\alpha_{\bar{z}}^\tau \mid b(\bar{z})] \right), \tag{118}$$

and

$$H_\tau = \mathbb{E}^{\mathcal{X}} \left[\mathrm{Cov}^{\mathcal{X}} \left(\Psi_{\bar{z}}, \alpha_{\bar{z}}^\tau \mid b(\bar{z}) \right) \right]. \tag{119}$$

Equation (115) is a partial-equilibrium derivative evaluated inside the HANK household block. The full general-equilibrium response additionally allows wages, hours, taxes, transfers, returns, relative prices, and the distribution to adjust.

A.5 Production and imported inputs

Domestic value added. Domestic producers use effective labor to produce the domestic value-added input:

$$X_t = Z_t N_t, \tag{120}$$

where X_t is real GDP and Z_t is aggregate productivity.

Let $P_{X,t}$ denote the nominal price of domestic value added and define

$$p_{X,t} \equiv \frac{P_{X,t}}{P_{H,t}}. \tag{121}$$

Domestic producers charge a constant flexible-price markup

$$\mu_p = \frac{\epsilon_p}{\epsilon_p - 1} \tag{122}$$

over nominal marginal cost. Since marginal cost is W_t/Z_t,

$$p_{X,t} = \mu_p \frac{w_t}{Z_t}. \tag{123}$$

Real dividends are

$$D_t = p_{X,t}X_t - w_tN_t - F_t, \tag{124}$$

where F_t is a fixed operating cost measured in units of the domestic final good. Using equation (123),

$$w_tN_t = \frac{p_{X,t}X_t}{\mu_p}, \tag{125}$$

and therefore

$$D_t = \left(1 - \frac{1}{\mu_p}\right) p_{X,t}X_t - F_t. \tag{126}$$

Final-good firms. Competitive final-good firms combine domestic value added $X_{H,t}$ and imported inputs $M_{I,t}$:

$$Y_t = \left[(1-\alpha_I)^{1/\nu_I} X_{H,t}^{\frac{\nu_I-1}{\nu_I}} + \alpha_I^{1/\nu_I} M_{I,t}^{\frac{\nu_I-1}{\nu_I}}\right]^{\frac{\nu_I}{\nu_I-1}}. \tag{127}$$

They solve

$$\min_{X_{H,t},M_{I,t}} \left\{p_{X,t}X_{H,t} + p^I_{M,t}M_{I,t}\right\} \tag{128}$$

subject to equation (127).

The first-order conditions imply

$$X_{H,t} = (1-\alpha_I)p_{X,t}^{-\nu_I}Y_t, \tag{129}$$

$$M_{I,t} = \alpha_I(p^I_{M,t})^{-\nu_I}Y_t. \tag{130}$$

Because $P_{H,t}$ is the final-good price, the unit-cost condition is

$$1 = \left[(1-\alpha_I)p_{X,t}^{1-\nu_I} + \alpha_I(p^I_{M,t})^{1-\nu_I}\right]^{\frac{1}{1-\nu_I}}. \tag{131}$$

The domestic value-added market clears:

$$X_{H,t} = X_t. \tag{132}$$

Equations (120), (129), and (132) determine the link between GDP, gross output, and labor.

Imported inputs are part of gross output but not domestic value added. They therefore do not enter GDP directly.

Exports. Foreign demand for the domestic final good is

$$EX_t = \overline{EX}q_t^{\eta_X}, \tag{133}$$

where $\eta_X > 0$. A real depreciation increases exports.

A.6 Wage setting

There is a continuum of differentiated labor services indexed by $j \in [0,1]$. Aggregate labor is

$$N_t = \left[\int_0^1 N_t(j)^{\frac{\epsilon_W - 1}{\epsilon_W}}\, dj\right]^{\frac{\epsilon_W}{\epsilon_W - 1}}. \tag{134}$$

Labor demand for variety j is

$$N_t(j) = \left(\frac{W_t(j)}{W_t}\right)^{-\epsilon_W} N_t. \tag{135}$$

The aggregate wage index satisfies

$$W_t^{1-\epsilon_W} = \int_0^1 W_t(j)^{1-\epsilon_W}\, dj. \tag{136}$$

Each union can reset its nominal wage with probability $1 - \theta_W$. A resetting union chooses W_t^*, knowing that this wage remains in place with probability θ_W^k at date $t+k$. Its labor demand at that date is

$$N_{t+k|t} = \left(\frac{W_t^*}{W_{t+k}}\right)^{-\epsilon_W} N_{t+k}. \tag{137}$$

The marginal utility benefit from one additional unit of real labor income is $(1 - \tau_t^\ell)\Lambda_t^e$. The reset-wage first-order condition is

$$\begin{aligned} 0 = \sum_{k=0}^{\infty} (\bar{\beta}_W \theta_W)^k \mathbb{E}_t \Bigg[N_{t+k|t} \Bigg\{ & (\epsilon_W - 1)(1 - \tau_{t+k}^\ell)\Lambda_{t+k}^e \frac{W_t^*}{P_{H,t+k}} \\ & - \epsilon_W v'(N_{t+k|t}) \Bigg\} \Bigg], \end{aligned} \tag{138}$$

where $\bar{\beta}_W$ is the discount factor used by the wage-setting union.

The desired gross wage markup is

$$\mu_W = \frac{\epsilon_W}{\epsilon_W - 1}. \tag{139}$$

Under flexible wages, the real wage satisfies

$$w_t^{\text{flex}} = \mu_W \frac{v'(N_t)}{(1 - \tau_t^\ell)\Lambda_t^e}. \tag{140}$$

The Calvo wage index evolves according to

$$W_t^{1-\epsilon_W} = \theta_W W_{t-1}^{1-\epsilon_W} + (1 - \theta_W)(W_t^*)^{1-\epsilon_W}. \tag{141}$$

Define the wage-gap variable

$$\widehat{g}_t^W \equiv \log\left(\frac{w_t^{\text{flex}}}{w_t}\right). \tag{142}$$

A first-order approximation around the zero-inflation stationary equilibrium gives

$$\pi_t^W = \bar{\beta}_W \mathbb{E}_t \pi_{t+1}^W + \kappa_W \widehat{g}_t^W, \tag{143}$$

where

$$\kappa_W = \frac{(1-\theta_W)(1-\bar{\beta}_W\theta_W)}{\theta_W(1+\epsilon_W\varphi)}. \tag{144}$$

Wage dispersion is second order around the symmetric stationary equilibrium and does not enter the first-order transition system.

A.7 Household cost-of-living indices and CPI

For a household with stationary composite consumption $\bar{c}_z$, define the compensated cost-of-living index as

$$P_{z,t}^{COL} = P_{H,t}\frac{\zeta_t + \mathcal{P}_{\xi,t}\bar{c}_z}{\bar{\zeta} + \bar{\mathcal{P}}_\xi\bar{c}_z}. \tag{145}$$

This index measures the expenditure required at date t to purchase the household's stationary utility level relative to its stationary expenditure.

Define stationary expenditure weights by

$$\omega_z^{\mathcal{X}} = \frac{\bar{x}_z\, d\bar{D}(z)}{\bar{\mathcal{X}}}. \tag{146}$$

The aggregate compensated CPI is

$$\log P_t^{CPI} = \int \log P_{z,t}^{COL}\, \omega_z^{\mathcal{X}}. \tag{147}$$

To first order,

$$d\log P_t^{CPI} = d\log P_{H,t} + \bar{s}_M d\log p_{M,t}^C, \tag{148}$$

where

$$\bar{s}_M = \frac{\bar{p}_M^C \bar{C}_M}{\bar{\mathcal{X}}} \tag{149}$$

is the stationary final-import expenditure share.

CPI inflation therefore satisfies

$$\pi_t^{CPI} = (1-\bar{s}_M)\pi_{H,t} + \bar{s}_M\pi_{M,t}^C, \tag{150}$$

where

$$\pi_{M,t}^C = \pi_{H,t} + \Delta\log p_{M,t}^C. \tag{151}$$

A.8 Monetary policy and asset prices

Monetary policy. The central bank follows

$$\log\left(\frac{1+i_t}{1+\bar{i}}\right) = \rho_i \log\left(\frac{1+i_{t-1}}{1+\bar{i}}\right) + (1-\rho_i)\left[\phi_\pi \log \Pi_t^{CPI} + \phi_X \log\left(\frac{X_t}{\bar{X}}\right)\right]. \tag{152}$$

The ex ante domestic real gross rate is

$$R_t = \frac{1+i_t}{\mathbb{E}_t \Pi_{H,t+1}}. \tag{153}$$

The realized real return on nominal government debt issued at $t-1$ is

$$R_t^B = \frac{1+i_{t-1}}{\Pi_{H,t}}. \tag{154}$$

Uncovered interest parity. Let R_t^* denote the foreign real gross interest rate. Perfect capital mobility implies

$$R_t = R_t^* \mathbb{E}_t\left(\frac{q_{t+1}}{q_t}\right). \tag{155}$$

In log-linear form,

$$\widehat{R}_t - \widehat{R}_t^* = \mathbb{E}_t \Delta \widehat{q}_{t+1}. \tag{156}$$

The realized return on foreign assets in domestic-good units is

$$R_t^F = R_{t-1}^* \frac{q_t}{q_{t-1}}. \tag{157}$$

A.9 Mutual fund

Households hold a single liquid claim issued by a competitive mutual fund. At the end of period t, the fund holds domestic equity J_t, government debt B_t, and net foreign assets NFA_t:

$$A_t = J_t + B_t + NFA_t. \tag{158}$$

Domestic equity is a claim to future dividends. In a deterministic transition, its value satisfies

$$J_t = \frac{D_{t+1} + J_{t+1}}{R_t}. \tag{159}$$

More generally,

$$J_t = \mathbb{E}_t\left[\frac{D_{t+1} + J_{t+1}}{R_t}\right]. \tag{160}$$

The realized payoff on the fund's beginning-of-period portfolio is

$$(1 + r_t^a)A_{t-1} = D_t + J_t + R_t^B B_{t-1} + R_t^F NFA_{t-1}. \tag{161}$$

Thus,

$$1 + r_t^a = \frac{D_t + J_t + R_t^B B_{t-1} + R_t^F NFA_{t-1}}{A_{t-1}}. \tag{162}$$

This return includes capital gains and losses on domestic equity. In particular, the period-zero household return incorporates the impact revaluation of J_t.

A.10 Fiscal policy

Tariff revenue is

$$\mathcal{R}_t^\tau = \tau_t^C q_t C_{M,t} + \tau_t^I q_t M_{I,t}. \tag{163}$$

Imports are valued at the border price because the tariff wedge is domestic revenue.

The government budget constraint is

$$B_t = R_t^B B_{t-1} + G_t + \mathcal{T}_t - \tau_t^\ell w_t N_t - \mathcal{R}_t^\tau, \tag{164}$$

where

$$\mathcal{T}_t = \int T_t(z)\, dD_t(z) \tag{165}$$

is aggregate transfers.

In the baseline,

$$\mathcal{T}_t = 0, \qquad G_t = \bar{G}. \tag{166}$$

Tariff revenue therefore reduces government borrowing on impact.

The labor-tax rule is

$$\tau_t^\ell - \bar{\tau}^\ell = \psi_B \frac{B_{t-1} - \bar{B}}{\bar{X}}, \qquad \psi_B > 0. \tag{167}$$

A small ψ_B implies gradual debt stabilization.

Under the uniform-rebate robustness,

$$T_t(z) = \mathcal{R}_t^\tau \qquad \text{for all } z, \tag{168}$$

after adjusting for unit population mass. Under revenue waste,

$$G_t^{\text{waste}} = \mathcal{R}_t^\tau, \tag{169}$$

and the expenditure does not enter household utility.

A.11 External balance

The trade balance in domestic-final-good units is

$$TB_t = EX_t - q_t \left(C_{M,t} + M_{I,t} \right). \tag{170}$$

Tariff-inclusive import prices do not enter the foreign import bill.

Net foreign assets evolve according to

$$NFA_t = R_t^F NFA_{t-1} + TB_t. \tag{171}$$

Equivalently, the current account is

$$CA_t \equiv NFA_t - NFA_{t-1} = (R_t^F - 1) NFA_{t-1} + TB_t. \tag{172}$$

A.12 Market clearing

The domestic final-good resource constraint is

$$Y_t = C_{H,t} + G_t + EX_t + F_t. \tag{173}$$

The fixed operating cost is a real use of the domestic final good.

The domestic value-added market clears according to

$$X_t = X_{H,t}. \tag{174}$$

The labor market clears according to

$$X_t = Z_t N_t. \tag{175}$$

The asset market clears according to

$$A_t = J_t + B_t + NFA_t. \tag{176}$$

Combining household, firm, government, fund, and external budget constraints gives Walras' law. One aggregate market-clearing condition is therefore redundant once all remaining equilibrium conditions hold.

A.13 Equilibrium definition

Given initial distributions and asset positions, exogenous foreign variables, productivity, and tariff processes, an equilibrium is a sequence

$$\left\{ \begin{array}{c} c_t(z),\, a_t(z),\, c_{H,t}(z),\, c_{M,t}(z),\, D_t(z), \\ N_t,\, X_t,\, Y_t,\, X_{H,t},\, M_{I,t},\, EX_t, \\ w_t,\, p_{X,t},\, p^C_{M,t},\, p^I_{M,t},\, q_t, \\ \Pi_{H,t},\, \Pi^W_t,\, \Pi^{CPI}_t,\, i_t, \\ D_t,\, J_t,\, A_t,\, B_t,\, NFA_t,\, \tau^\ell_t \end{array} \right\}_{t\geq 0} \tag{177}$$

such that:

1. household policies solve equations (81)–(83);
2. the household distribution satisfies equation (94);
3. conditional consumption demands satisfy equations (75) and (76);
4. domestic and final-good firms satisfy their cost-minimization and pricing conditions;
5. wage setters satisfy equations (138) and (141);
6. monetary and fiscal authorities satisfy their policy rules and budget constraints;
7. the mutual fund satisfies its balance sheet and return conditions;
8. uncovered interest parity and the external budget constraint hold;
9. all goods, labor, and asset markets clear.

A.14 Stationary equilibrium

A stationary equilibrium has constant real quantities and zero domestic and wage inflation:

$$\bar{\Pi}_H = \bar{\Pi}^W = \bar{\Pi}^{CPI} = 1. \tag{178}$$

The stationary real rate satisfies

$$\bar{R} = 1 + \bar{i}. \tag{179}$$

The real exchange rate is normalized to

$$\bar{q} = 1. \tag{180}$$

For each permanent type (β, ξ), stationary household policies solve

$$\bar{V}(a, e; \beta, \xi) = \max_{c,a'} \quad u(c) - v(\bar{N}) + \beta \sum_{e'} \Pi_e(e, e')\bar{V}(a', e'; \beta, \xi) \tag{181}$$

$$\text{s.t.} \quad \bar{\zeta} + \bar{\mathcal{P}}_\xi c + a' = (1 + \bar{r}^a)a + (1 - \bar{\tau}^\ell)\bar{w}e\bar{N} + \bar{T}(e, \beta, \xi). \tag{182}$$

The stationary distribution satisfies

$$\bar{D} = \mathcal{T}(\bar{g}_a, \Pi_e)\bar{D}, \tag{183}$$

where $\mathcal{T}$ is the distribution-transition operator.

The stationary production block satisfies

$$\bar{X} = \bar{Z}\bar{N}, \tag{184}$$

$$\bar{p}_X = \mu_p \frac{\bar{w}}{\bar{Z}}, \tag{185}$$

$$1 = \left[(1-\alpha_I)\bar{p}_X^{1-\nu_I} + \alpha_I(\bar{p}_M^I)^{1-\nu_I}\right]^{\frac{1}{1-\nu_I}}, \tag{186}$$

$$\bar{X} = (1-\alpha_I)\bar{p}_X^{-\nu_I}\bar{Y}, \tag{187}$$

$$\bar{M}_I = \alpha_I(\bar{p}_M^I)^{-\nu_I}\bar{Y}. \tag{188}$$

The stationary fiscal and external positions satisfy

$$\bar{B} = \bar{R}\bar{B} + \bar{G} + \bar{\mathcal{T}} - \bar{\tau}^\ell \bar{w}\bar{N} - \bar{\mathcal{R}}^\tau, \tag{189}$$

$$(1-\bar{R}^F)\overline{NFA} = \overline{TB}. \tag{190}$$

The quantitative baseline normalizes

$$\bar{B} = 0, \qquad \overline{NFA} = 0, \qquad \overline{TB} = 0. \tag{191}$$

The fixed operating cost is chosen consistently with stationary dividends and the domestic resource constraint:

$$\bar{F} = \bar{Y} - \bar{C}_H - \bar{G} - \overline{EX}. \tag{192}$$

Equivalently,

$$\bar{F} = \left(1 - \frac{1}{\mu_p}\right)\bar{p}_X\bar{X} - \bar{D}. \tag{193}$$

The two expressions must coincide.

A.15 The three exposure economies

The quantitative analysis compares three household demand systems.

Common-basket economy. The common-basket economy removes both sources of exposure heterogeneity:

$$\bar{c}_H = \bar{c}_M = 0, \qquad \omega_\xi = \omega^{\text{common}} \quad \text{for all } \xi. \tag{194}$$

The common weight is chosen to reproduce the baseline aggregate final import share. At given prices, every household has the same import expenditure share.

Vertical-exposure economy. The vertical economy retains the Stone–Geary subsistence quantities but removes basket-type dispersion:

$$\bar{c}_H > 0, \qquad \bar{c}_M > 0, \qquad \omega_\xi = \omega^{\text{vertical}} \quad \text{for all } \xi. \tag{195}$$

Import shares vary with household expenditure but not with basket type conditional on expenditure.

Full-exposure economy. The full economy retains both non-homotheticity and basket-type heterogeneity:

$$\bar{c}_H > 0, \qquad \bar{c}_M > 0, \qquad \omega_\xi \in \left\{\omega_1, \ldots, \omega_{N_\xi}\right\}. \tag{196}$$

The distribution of ω_ξ is calibrated to match residual exposure dispersion within expenditure groups.

The aggregate final import share is matched across the three economies. Each economy is resolved separately, producing its own stationary distribution, policy functions, and household Jacobians.

A.16 First-order aggregate system

Let a hat denote a log deviation from stationary equilibrium. The log-linear unit-cost condition is

$$0 = (1 - \bar{s}_I)\widehat{p}_{X,t} + \bar{s}_I \widehat{p}^I_{M,t}, \tag{197}$$

where

$$\bar{s}_I = \frac{\bar{p}^I_M \bar{M}_I}{\bar{Y}}. \tag{198}$$

Flexible product pricing implies

$$\widehat{p}_{X,t} = \widehat{w}_t - \widehat{Z}_t. \tag{199}$$

Value added satisfies

$$\widehat{X}_t = \widehat{Z}_t + \widehat{N}_t. \tag{200}$$

Conditional input demands satisfy

$$\widehat{X}_t = \widehat{Y}_t - \nu_I \widehat{p}_{X,t}, \tag{201}$$

$$\widehat{M}_{I,t} = \widehat{Y}_t - \nu_I \widehat{p}^I_{M,t}. \tag{202}$$

Import prices satisfy

$$\widehat{p}^C_{M,t} = \widehat{q}_t + \vartheta^C_t, \tag{203}$$

$$\widehat{p}^I_{M,t} = \widehat{q}_t + \vartheta^I_t. \tag{204}$$

Exports satisfy

$$\widehat{EX}_t = \eta_X \widehat{q}_t. \tag{205}$$

The wage Phillips curve is

$$\pi_t^W = \bar{\beta}_W \mathbb{E}_t \pi_{t+1}^W + \kappa_W \left(\widehat{w}_t^{\text{flex}} - \widehat{w}_t \right). \tag{206}$$

The flexible real wage satisfies

$$\widehat{w}_t^{\text{flex}} = \varphi \widehat{N}_t - \widehat{\Lambda}_t^e + \frac{\bar{\tau}^\ell}{1 - \bar{\tau}^\ell} \widehat{\tau}_t^\ell. \tag{207}$$

Domestic inflation is

$$\pi_{H,t} = \pi_t^W - \Delta \widehat{w}_t. \tag{208}$$

CPI inflation is

$$\pi_t^{CPI} = \pi_{H,t} + \bar{s}_M \Delta \widehat{p}_{M,t}^C. \tag{209}$$

The log-linear Taylor rule is

$$\widehat{i}_t = \rho_i \widehat{i}_{t-1} + (1 - \rho_i) \left(\phi_\pi \pi_t^{CPI} + \phi_X \widehat{X}_t \right). \tag{210}$$

The ex ante real rate is

$$\widehat{R}_t = \widehat{i}_t - \mathbb{E}_t \pi_{H,t+1}. \tag{211}$$

UIP is

$$\widehat{R}_t - \widehat{R}_t^* = \mathbb{E}_t \left(\widehat{q}_{t+1} - \widehat{q}_t \right). \tag{212}$$

The labor-tax rule is

$$\widehat{\tau}_t^\ell = \psi_B \frac{B_{t-1} - \bar{B}}{\bar{X}}. \tag{213}$$

The government budget and fund balance sheet are linearized in levels when their stationary values are zero.

A.17 Sequence-space solution

The household block maps sequences of aggregate inputs into sequences of household aggregates. Let

$$\mathbf{u} = \left\{ r_t^a,\, w_t,\, N_t,\, \tau_t^\ell,\, T_t,\, p_{M,t}^C \right\}_{t=0}^{T-1} \tag{214}$$

denote household inputs and let

$$\mathbf{h} = \{ A_t,\, C_t,\, \mathcal{X}_t,\, C_{H,t},\, C_{M,t},\, \Lambda_t^e \}_{t=0}^{T-1} \tag{215}$$

denote household outputs.

Around the stationary equilibrium,

$$d\mathbf{h} = \mathbf{J}^H d\mathbf{u}, \tag{216}$$

where $\mathbf{J}^H$ contains the sequence-space Jacobians of the household block.

The remaining equilibrium conditions can be stacked as

$$\mathcal{F}(\mathbf{y}, \mathbf{h}, \mathbf{z}) = 0, \tag{217}$$

where $\mathbf{y}$ collects endogenous aggregate sequences and $\mathbf{z}$ collects exogenous shocks.

Substituting equation (216) into the linearization of equation (217) gives

$$\left[\mathcal{F}_{\mathbf{y}} + \mathcal{F}_{\mathbf{h}}\mathbf{J}^H_{\mathbf{y}}\right] d\mathbf{y} = -\left[\mathcal{F}_{\mathbf{z}} + \mathcal{F}_{\mathbf{h}}\mathbf{J}^H_{\mathbf{z}}\right] d\mathbf{z}. \tag{218}$$

The equilibrium transition is

$$d\mathbf{y} = -\left[\mathcal{F}_{\mathbf{y}} + \mathcal{F}_{\mathbf{h}}\mathbf{J}^H_{\mathbf{y}}\right]^{-1} \left[\mathcal{F}_{\mathbf{z}} + \mathcal{F}_{\mathbf{h}}\mathbf{J}^H_{\mathbf{z}}\right] d\mathbf{z}. \tag{219}$$

Separate household Jacobians are computed for the common-basket, vertical-exposure, and full-exposure economies. The general-equilibrium blocks are otherwise held fixed.

B Additional model variants

B.1 Delayed adjustment of consumption baskets

The baseline allocation between domestic and imported final goods is frictionless. As a robustness exercise, let

$$b_{i,t} \equiv \log\left(\frac{c_{M,i,t} - \bar{c}_M}{c_{H,i,t} - \bar{c}_H}\right) \tag{220}$$

denote the log composition of the supernumerary basket. The frictionless target is

$$b^*_{i,t} = \log\left(\frac{\omega_{\xi_i}}{1 - \omega_{\xi_i}}\right) - \eta_C \log p^C_{M,t}. \tag{221}$$

Conditional on total supernumerary expenditure, the household solves

$$\min_{\{b_{i,t+s}\}_{s\geq 0}} \mathbb{E}_t \sum_{s=0}^{\infty} \bar{\beta}_b^s \left[\frac{1}{2}(b_{i,t+s} - b^*_{i,t+s})^2 + \frac{\lambda_b}{2}(b_{i,t+s} - b_{i,t+s-1})^2\right]. \tag{222}$$

The first-order condition is

$$\left[1 + \lambda_b(1 + \bar{\beta}_b)\right] b_{i,t} - \lambda_b b_{i,t-1} - \bar{\beta}_b \lambda_b \mathbb{E}_t b_{i,t+1} = b^*_{i,t}. \tag{223}$$

The frictionless benchmark is recovered at $\lambda_b = 0$.

Given supernumerary expenditure $x_{i,t}^S$ and basket composition $b_{i,t}$,

$$c_{H,i,t} - \bar{c}_H = \frac{x_{i,t}^S}{1 + p_{M,t}^C \exp(b_{i,t})}, \tag{224}$$

$$c_{M,i,t} - \bar{c}_M = \frac{\exp(b_{i,t}) x_{i,t}^S}{1 + p_{M,t}^C \exp(b_{i,t})}. \tag{225}$$

This block replaces the frictionless conditional-demand equations. It does not add a second import-demand equation.

B.2 Sticky domestic product prices

As a robustness exercise, domestic value-added producers reset prices with probability $1 - \theta_p$. Let mc_t denote real marginal cost:

$$mc_t = \frac{w_t}{Z_t p_{X,t}}. \tag{226}$$

The first-order domestic-price Phillips curve is

$$\pi_{X,t} = \bar{\beta}_p \mathbb{E}_t \pi_{X,t+1} + \kappa_p \widehat{mc}_t, \tag{227}$$

where

$$\kappa_p = \frac{(1-\theta_p)(1-\bar{\beta}_p\theta_p)}{\theta_p}. \tag{228}$$

The baseline is recovered as $\theta_p \to 0$, in which case

$$p_{X,t} = \mu_p \frac{w_t}{Z_t}. \tag{229}$$

C Robustness

This appendix tests the main results along four dimensions. First, it separates tariffs on final consumption goods from tariffs on imported inputs. Second, it varies the tariff process, monetary policy, nominal rigidity, production, household demand, and the fiscal and external closures. Third, it changes the sorting between permanent basket types and household patience. Finally, it verifies that the results are stable to the resolution of the household asset grid.

Two conclusions are robust. The level of the aggregate response depends on the production structure and the policy closure. The difference between the full-exposure and common-basket economies is much more stable: heterogeneous exposure makes the impact output response more negative in every specification considered.

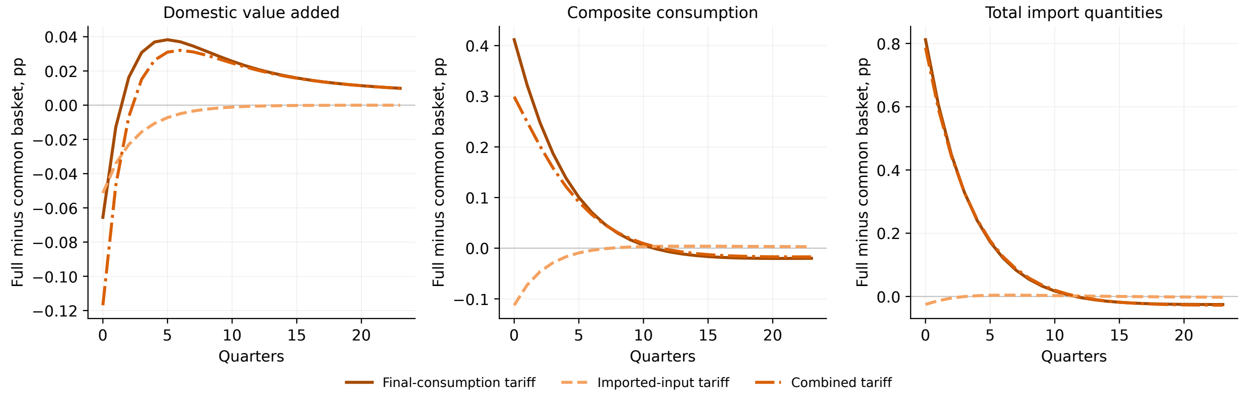

Figure 6: Exposure gaps across tariff margins. The figure reports the full-exposure response minus the common-basket response under a tariff on final consumption imports, a tariff on imported inputs, and a common tariff on both uses. Differences are expressed in percentage points.

C.1 Final-goods and imported-input tariffs

Figure 6 decomposes the difference between the full-exposure and common-basket economies by tariff margin.

The household channel. A tariff on final consumption imports makes impact domestic value added approximately 0.066 percentage points lower in the full-exposure economy. At the same time, composite consumption falls around 0.412 percentage points less and total import quantities fall around 0.812 percentage points less.

These responses reflect weaker expenditure switching. Necessity-like import demand makes highly exposed households reduce imported consumption less aggressively. They preserve more imports but redirect less expenditure toward domestic goods. Domestic value added therefore falls more even though aggregate composite consumption falls less.

The production channel. A tariff on imported inputs makes impact domestic value added approximately 0.051 percentage points lower in the full-exposure economy. It also makes composite consumption slightly more contractionary. This margin does not directly change household import prices, but it affects households through labor income, asset returns, and the equilibrium response of domestic production.

Because the model is solved to first order, the response to the common tariff is the sum of the two separate experiments. Both channels contribute to the additional fall in domestic value added. Most of the difference in household consumption and import quantities, however, comes from the final-consumption tariff.

The gaps are concentrated in the first few quarters. Exposure heterogeneity changes the impact and early transition but has little effect once the tariff wedge dissipates.

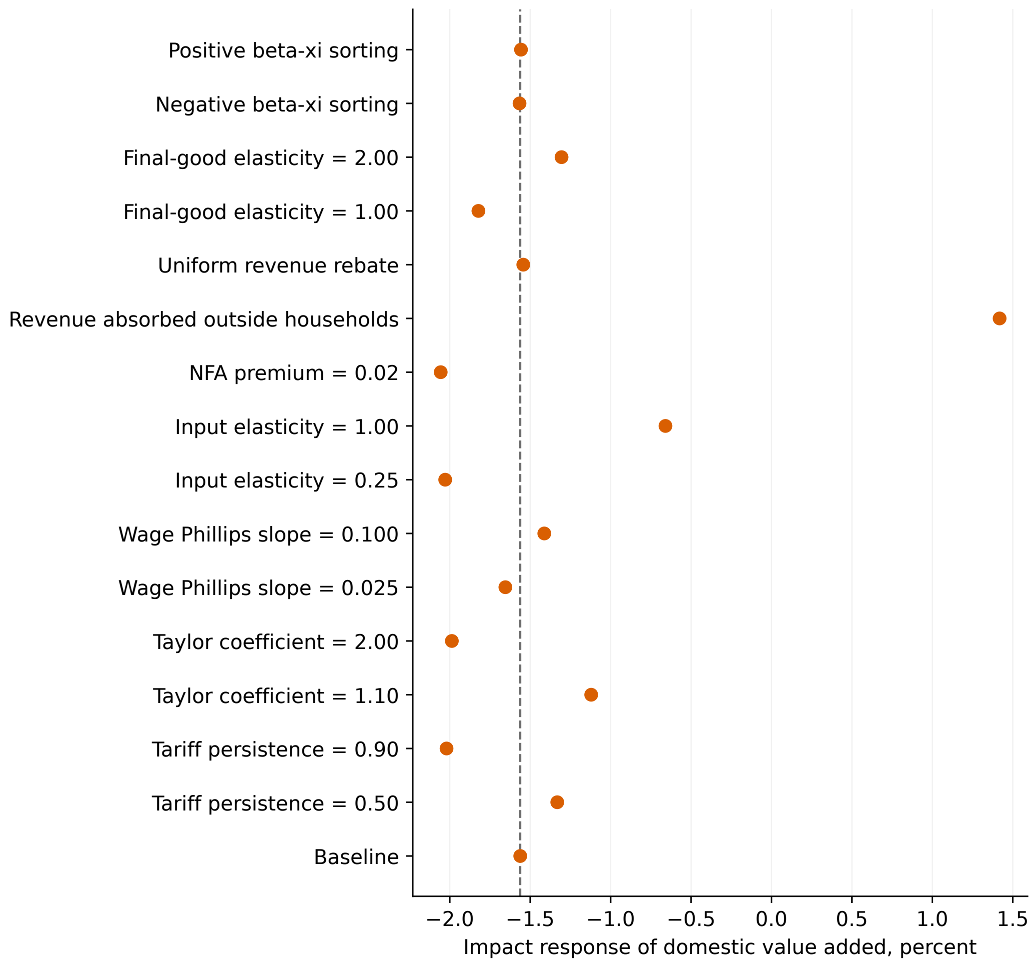

Figure 7: Robustness of the impact output response. Each marker reports the impact response of domestic value added in the full-exposure economy. The dashed vertical line denotes the baseline response.

C.2 Alternative calibrations and policy closures

Figure 7 reports the impact output response in the full-exposure economy under alternative specifications. The dashed line denotes the baseline.

Table 3 reports the corresponding impact responses and the output difference relative to the common-basket economy.

Tariff persistence and monetary policy. A more persistent tariff produces a larger output contraction, while a less persistent tariff attenuates it. A stronger CPI response also deepens the fall in activity. These parameters materially change the level of the aggregate response but have little effect on the full–common difference.

The same pattern holds for wage rigidity. Greater wage rigidity increases the contraction and modestly enlarges the exposure gap. More flexible wages reduce both.

Production and external adjustment. The elasticity between domestic value added and imported inputs is important for the level of output. When inputs are difficult to replace, the tariff generates a larger domestic contraction. With a unit elasticity, firms substitute more strongly and the output loss becomes substantially smaller, although imports fall more.

Introducing a debt-elastic NFA premium increases the output contraction and reduces the full–common gap. External adjustment therefore affects the quantitative size of the household mechanism, but not its sign.

Fiscal closure. Fiscal policy is the most consequential robustness dimension. A uniform rebate substantially cushions household consumption but leaves the impact output contraction close to the baseline. By contrast, the extreme closure in which tariff revenue is absorbed outside households and does not reduce government debt reverses the impact output response, even though household consumption still falls.

This sign reversal is specific to the treatment of tariff revenue and should not be read as a central prediction. It shows that the aggregate level effect of a tariff cannot be separated from the government balance-sheet response. Importantly, the full-exposure economy still produces lower output than the common-basket economy under every fiscal closure.

Household substitution. Changing the elasticity between domestic and imported final goods alters both expenditure switching and the aggregate contraction. A lower elasticity produces a larger output decline but a smaller full–common gap. A higher elasticity produces a smaller aggregate contraction but a larger exposure gap.

Across all specifications in Table 3, the full-exposure impact output response is between 0.055 and 0.153 percentage points more negative than in the common-basket economy. The central aggregate result is therefore not driven by a single calibration or policy rule.

C.3 Sorting between exposure and spending responses

The baseline assumes that permanent basket type and discount-factor type are independent. This does not eliminate sorting between exposure and iMPCs because Stone–Geary demand generates an endogenous expenditure gradient. Table 4 changes only the joint distribution of permanent types while preserving their marginal distributions.

Positive sorting raises the exposure–iMPC correlation and makes both the vertical and horizontal components more contractionary. The total direct effect increases from 0.407 percent under negative sorting to 0.444 percent under positive sorting.

Under negative sorting, the horizontal component becomes positive and partly offsets the vertical gradient. Under positive sorting, it reinforces that gradient. The result follows the sufficient statistic directly: horizontal exposure heterogeneity amplifies aggregate demand only when high-exposure baskets are assigned to high-iMPC households.

The baseline remains between these two cases. Even under independence, the direct effect is larger than the average-exposure benchmark because resources jointly determine liquidity and Stone–Geary import shares.

C.4 Numerical accuracy

Table 5 recomputes the stationary household block and the direct-effect decomposition using alternative asset-grid resolutions.

The quarterly MPC, aggregate import share, exposure–iMPC correlation, and direct expenditure effect are essentially unchanged across grids. The small movement in aggregate assets reflects approximation at the upper part of the wealth distribution and does not affect the mechanism.

Across all general-equilibrium robustness exercises, the maximum residual of the stacked equilibrium system remains below 3×10^{-13}. The numerical implementation also preserves the direct-effect decomposition exactly and is stable to alternative equilibrium closures and solution horizons.

Table 3: Alternative calibrations and policy closures

Specification	Output	Consumption	Imports	CPI inflation	Full–common output
Panel A: Baseline, tariff process, and nominal block					
Baseline	−1.56	−4.94	−7.63	4.17	−0.117
Tariff persistence: $\rho_\tau = 0.50$	−1.33	−4.78	−7.58	4.37	−0.127
Tariff persistence: $\rho_\tau = 0.90$	−2.02	−5.23	−7.70	3.70	−0.102
Taylor coefficient: $\phi_\pi = 1.10$	−1.12	−4.64	−7.55	4.52	−0.121
Taylor coefficient: $\phi_\pi = 2.00$	−1.99	−5.22	−7.70	3.85	−0.116
Wage Phillips slope: $\kappa_W = 0.025$	−1.65	−5.00	−7.64	4.23	−0.138
Wage Phillips slope: $\kappa_W = 0.100$	−1.41	−4.84	−7.60	4.09	−0.090
Panel B: Production, external, and fiscal blocks					
Input elasticity: $\nu_I = 0.25$	−2.03	−5.05	−6.68	4.14	−0.122
Input elasticity: $\nu_I = 1.00$	−0.66	−4.74	−9.44	4.24	−0.108
NFA premium: $\phi_{NFA} = 0.02$	−2.06	−4.06	−7.03	3.51	−0.055
Revenue absorbed outside households	1.42	−4.58	−5.46	4.38	−0.108
Uniform revenue rebate	−1.54	−3.34	−6.72	4.33	−0.080
Panel C: Household demand					
Final-good elasticity: $\eta_C = 1.00$	−1.82	−5.00	−6.96	4.15	−0.080
Final-good elasticity: $\eta_C = 2.00$	−1.31	−4.88	−8.28	4.19	−0.153

Notes: Output, consumption, and total imports are impact percentage deviations in the full-exposure economy. CPI inflation and the full–common output difference are in percentage points. The baseline tariff is applied jointly to final consumption goods and imported inputs.

Table 4: Exposure sorting and the direct expenditure effect

Sorting	Exposure–iMPC correlation	Average exposure	Vertical	Horizontal	Total
Negative	0.247	−0.383	−0.031	0.007	−0.407
Independent	0.439	−0.382	−0.040	−0.003	−0.425
Positive	0.622	−0.381	−0.049	−0.014	−0.444

Notes: The decomposition is evaluated on impact for the baseline tariff shock. Components are expressed in percent of aggregate household expenditure. Positive sorting pairs low-discount-factor, high-iMPC households with more import-intensive basket types. Negative sorting reverses this assignment.

Table 5: Asset-grid robustness

Grid points	Assets / GDP	Quarterly MPC	Four-quarter MPC	Import share	Direct effect	Exposure–iMPC correlation
120	10.0826	0.19996	0.28262	0.29996	−0.42482	0.43924
160	10.0000	0.20000	0.28292	0.30000	−0.42526	0.43926
200	9.9529	0.20005	0.28318	0.30002	−0.42558	0.43933

Notes: The baseline uses 160 asset-grid points. The direct effect is expressed in percent of aggregate household expenditure.